



University of Colorado **Anschutz Medical Campus**

Introduction to Alpine

By Kevin Fotso
07/21/2026



Introduction:

This workshop will cover the following topics:

- What is Alpine and how to log into the Alpine system.
- Alpine filesystems and storage options
- Alpine data transfer options
- Alpine scheduler



Introduction:

This workshop will cover the following topics:

- *Alpine modules and software stack*
- *Alpine interactive and batch jobs*
- *Alpine resource monitoring*



Audience:

Users who are either new to Alpine or hoping to gain more Alpine experience



University of Colorado
Anschutz Medical Campus

Alpine requirements

Reminder! Alpine cannot be used on sensitive data!



- No HIPAA or Protected Health Information (PHI)
- No General Data Protection Regulation (GDPR)
 - No CMMC
 - No FERPA
- No data that requires special security and compliance



FERPA
Family Educational
Rights & Privacy Act



All data on Alpine must be 100% de-identified and make sure it is compliant with your IRB and/or grant funding agency

HPC

Benefits of HPC – Resource availability



Standard Laptop
~8GB RAM,
500GB hard drive,
Dual-core processor



**Standard CPU Node on
Alpine (amilan nodes)**
64 cores, ~3.74GB/core
=240GB RAM

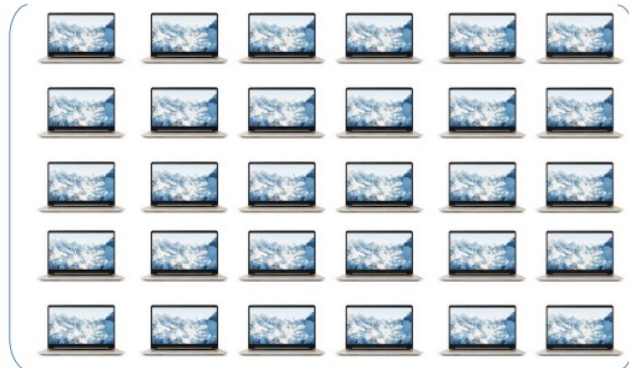


**Alpine High
Performance Compute
Cluster**

456 compute nodes for a
total of 28,144 cores*

Alpine also has special
nodes such as GPUs
and high memory nodes
which contain up to 2T of RAM.

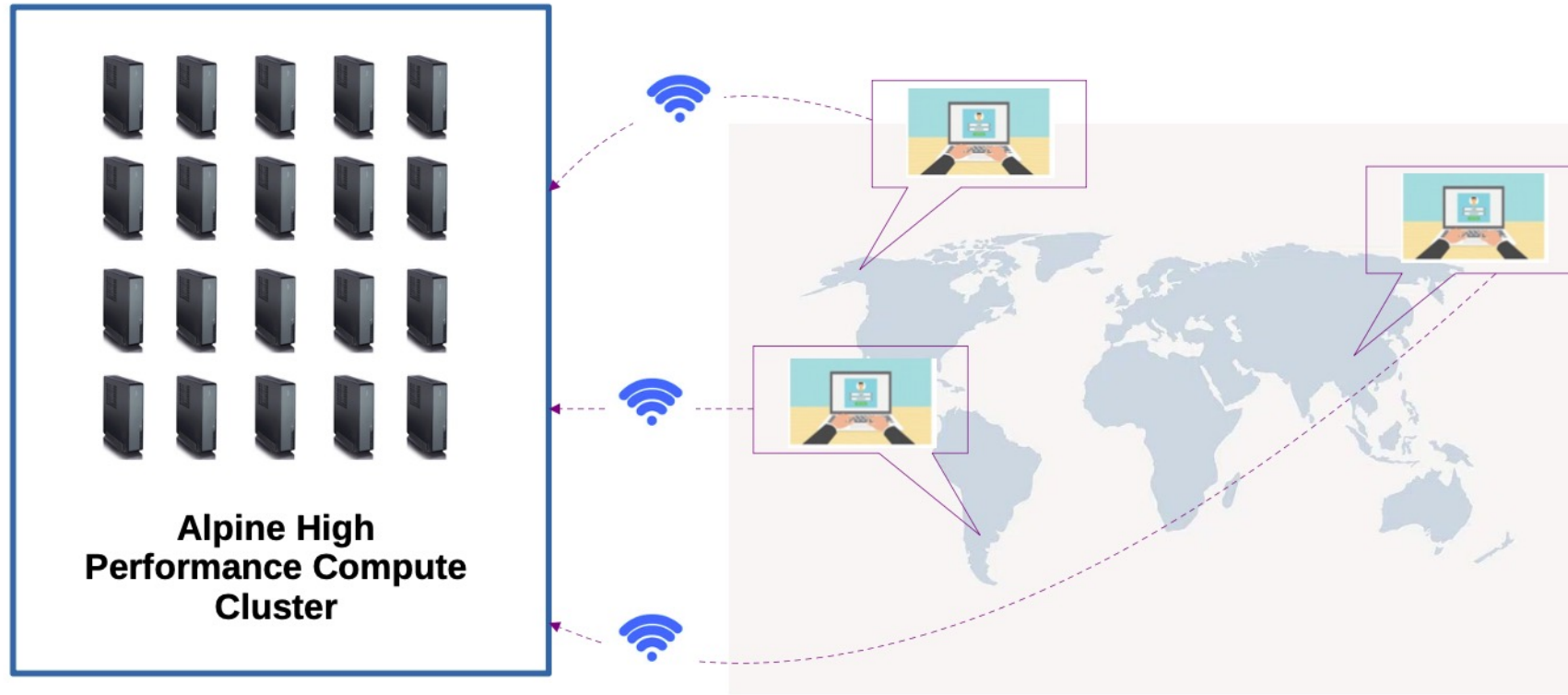
30 standard
laptops = 1 CPU
node on Alpine!



* includes GPU & high mem

HPC

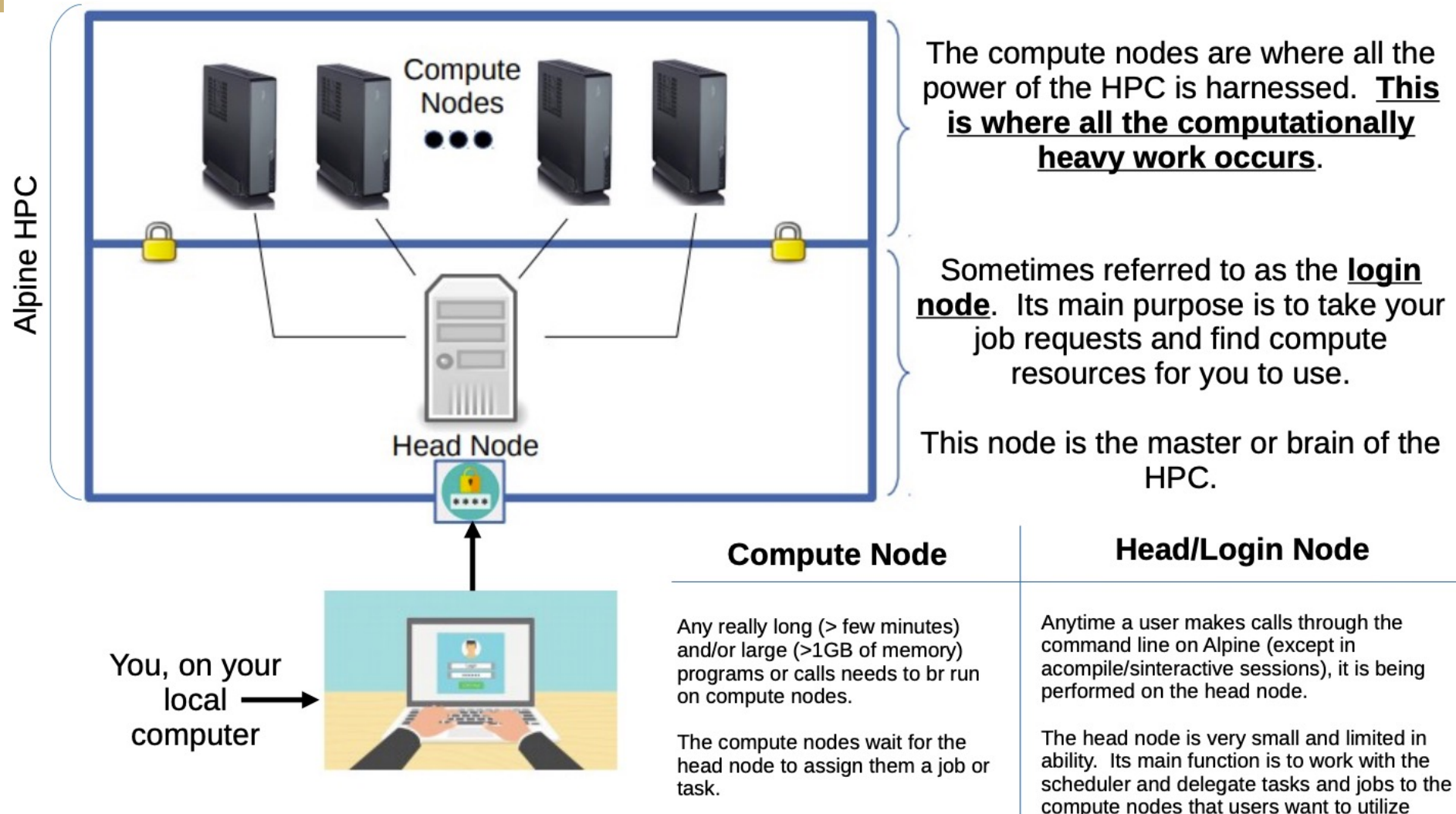
Benefits of HPC – Remote computation



Once you submit your job to Alpine or an HPC, you can turn off your computer. At this point, all the data, commands, scripts, etc... are copied over to the Boulder side where the power to the server is always on, therefore **turning off your computer after you submit your job is not at all tied to your local computer power.**

HPC

Anatomy of HPC



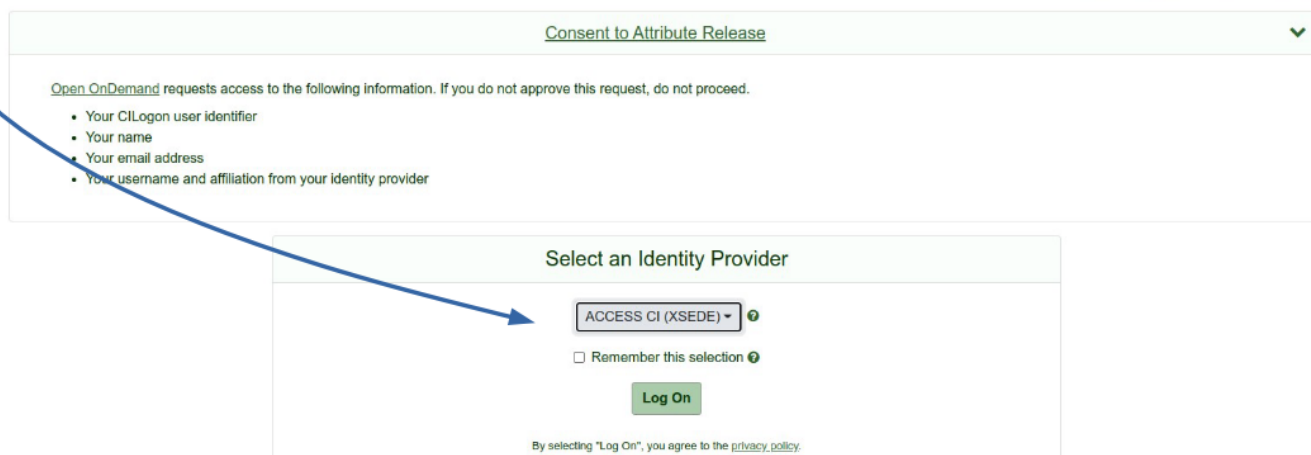
How to access Alpine

First let's login to Alpine!



1 Go to <https://ondemand-rmacc.rc.colorado.edu>

2 It should redirect you to CILogon which is how you authenticate your Alpine session. Make sure you select "ACCESS CI (XSEDE)" as your identity provide and then press "Log On"



[Consent to Attribute Release](#)

[Open OnDemand](#) requests access to the following information. If you do not approve this request, do not proceed.

- Your CILogon user identifier
- Your name
- Your email address
- Your username and affiliation from your identity provider

Select an Identity Provider

ACCESS CI (XSEDE)

☐ Remember this selection

Log On

By selecting "Log On", you agree to the [privacy policy](#).

6

How to access Alpine

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Select an Identity Provider

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☐ Remember this selection

Log On

By selecting "Log On", you agree to the [privacy policy](#).

Note: Please do not use the Firefox browser to access ondemand at this time.

How to access Alpine

First let's login to Alpine!



3

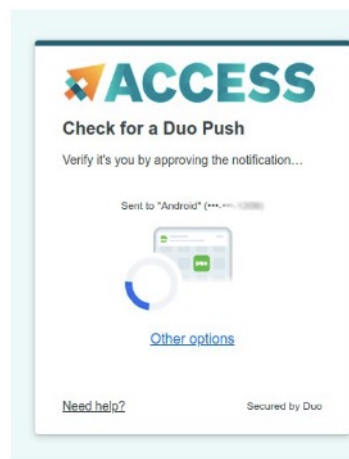
Next it will take you to this page where you will put in your ACCESS ID and ACCESS password and press Login. This is NOT your CU Anschutz ID!!

ACCESS ID
and ACCESS
password **not**
CU Anschutz
credentials!!

The screenshot shows the ACCESS login page. On the left, there's a form titled "Login to CILogon" with fields for "ACCESS Username" and "ACCESS Password", and a "Login" button. On the right, there's a CILogon logo and text explaining its purpose. Below the logo, there's a warning triangle icon and text: "If you had an XSEDE account, please enter your XSEDE username and password for ACCESS login". Below this, there are three links: "Register for an ACCESS Account", "Forgot your password?", and "Need Help?".

4

This will prompt a DUO MFA push to your phone or whichever way you have DUO set up on your phone to authenticate. Accept the push sent to your device.



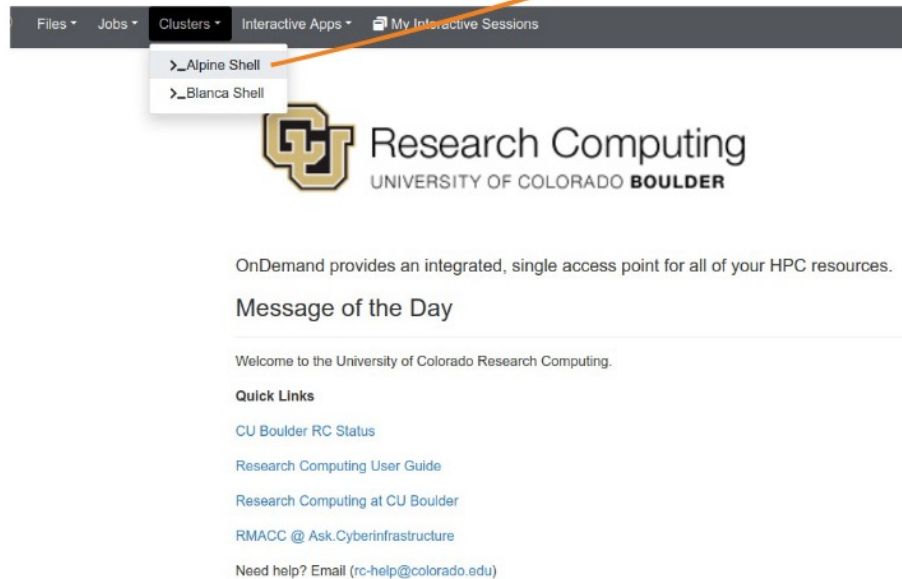
How to access Alpine

First let's login to Alpine!



5

This will take to the official CURC page. Let's select the Alpine terminal



Make sure that you can access Alpine

First let's login to Alpine!



6

This will log you into the head node of Alpine. It will always default you to your home directory.

```
Host: login-cil.rc.int.colorado.edu
Welcome to University of Colorado Boulder Research Computing!

Full documentation is available in our user guide at
https://www.rc.colorado.edu/support/user-guide. If you have a question
that's not answered there, contact us at rc-help@colorado.edu.

A number of directories have been created for you already:

* `/home/$USER`, your home directory
* `/projects/$USER`, your project directory

Run the command `module avail` to see a list of available software.

To prevent this README from being displayed at login, edit your
`.bash_profile` or `.login` files.
Welcome to CU-Boulder Research Computing.

* Website http://colorado.edu/rc
* Questions? rc-help@colorado.edu
* Subscribe to system announcements: https://curc.statuspage.io/
* Please type rc-help for the Acceptable Use Policy and a short help page.

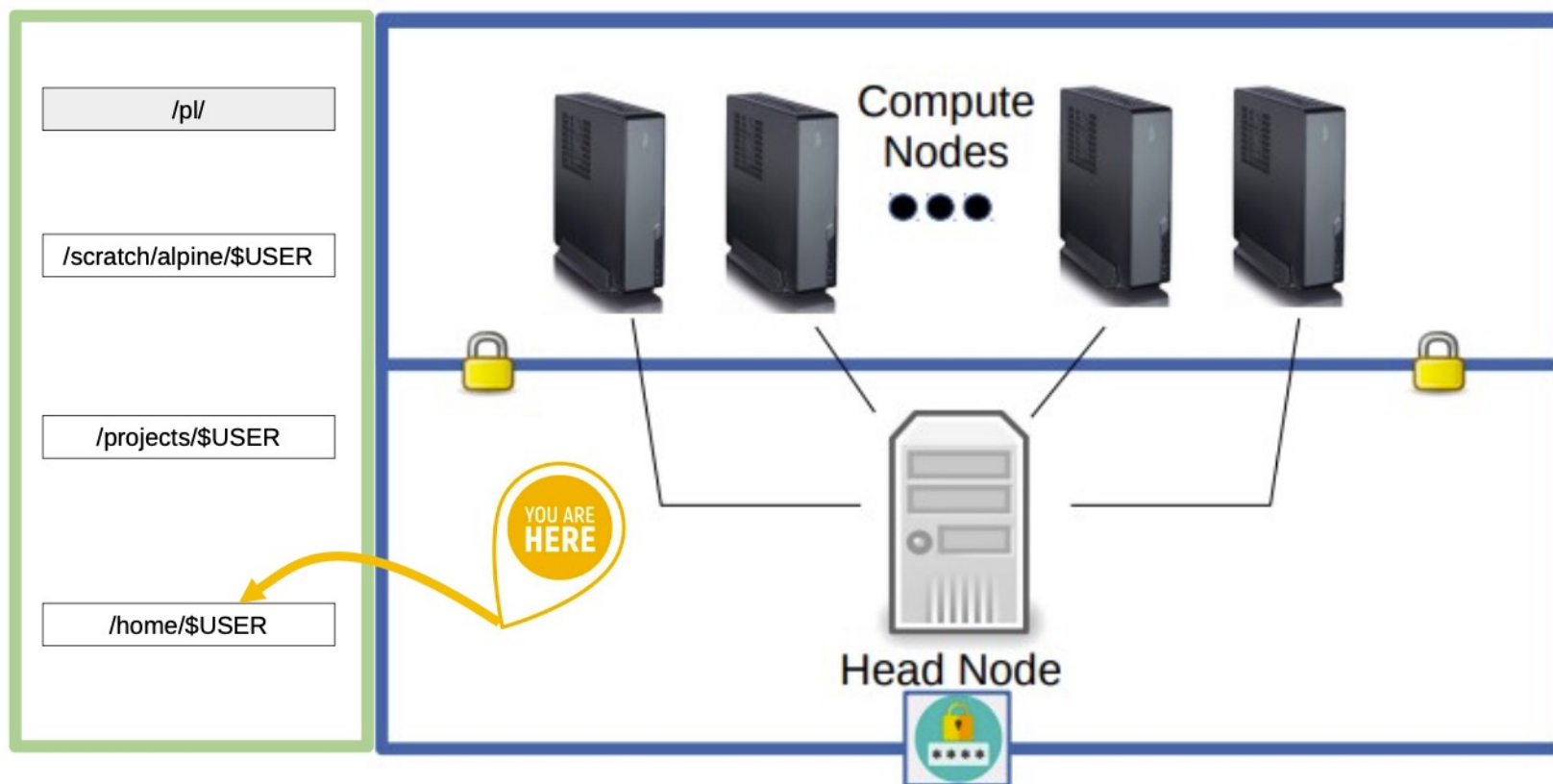
You are using login node: login-cil

Users who had jobs in the queue prior to the planned maintenance should check
to confirm these jobs are still queued. Some jobs, particularly those schedule
since midnight today (Wed June 7), may have been canceled during the
maintenance period.
@xsede.org@login-cil @xsede.org]$
```

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Storage

All 4 storage directories you have for storage are accessible by both the head/login node and the compute nodes



Every time you login to Alpine, **you will always be located in your /home/\$USER directory on the head/login node**



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Source: [https://github.com/kf-cuanschutz/CU-Anschutz-HPC-documentation/blob/main/Workshops/Alpine Noob Introduction to HPC and Alpine-120723.pdf](https://github.com/kf-cuanschutz/CU-Anschutz-HPC-documentation/blob/main/Workshops/Alpine%20Noob%20Introduction%20to%20HPC%20and%20Alpine-120723.pdf)

Filesystems

Alpine HPC RoadMap

/home/\$USER

/projects/\$USER

/scratch/alpine/\$USER

/pl/

Every time you login to Alpine, **you will always be located in your /home/\$USER directory**

Filesystems

Alpine HPC RoadMap

/home/\$USER

/projects/\$USER

/scratch/alpine/\$USER

/pl/

Every time you login to Alpine, **you will always be located in your /home/\$USER directory**

It doesn't matter if you were in your scratch space when you logged out or where you were in any prior session; Alpine does not remember where you were last time, so it always puts you in your /home/\$USER directory space

Filesystems

```
/home/$USER
```

/home/\$USER is very, very small!
2GB total

For most users, you never want to do anything to or in home; this is where most your configuration files are located, such as your .bashrc, .bash_profile, .condarc, etc...

The only time you will do anything in your home space is if you want to modify these hidden configuration files

Do not place any data transfer here or store any data here!

Filesystems

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Do not place any data transfer here or store any data here!

Do not change any file under /home/\$USER/.ssh unless you really know what you are doing!

Filesystems

```
/home/$USER
```

/home/\$USER is very, very small!
2GB total

For most users, you never want to do anything to or in home; this is where most your configuration files are located, such as your .bashrc, .bash_profile, .condarc, etc...

The only time you will do anything in your home space is if you want to modify these hidden configuration files

Do not place any data transfer here or store any data here!

If you need to set up github with Alpine please refer to this guide:
https://github.com/ResearchComputing/git_github_in_depth_short_course

Filesystems

/home/\$USER

/projects/\$USER

/scratch/alpine/\$USER

/pl/

Let's change directories into our projects directory...



```
cd /projects/$USER
```

Filesystems

```
/projects/$USER
```

This directory is slightly larger, at 250GB of space.

This should be used for more permanent storage of smaller files.

I recommend any local software installs (i.e. software you install yourself because it is not installed on Alpine) get installed here.

Additionally, all of your job scripts and code should be stored here

If small enough, reference data should be installed here; reference data meaning data that is used over and over again. In bioinformatics, this usually means genome index files, reference genome files (.fasta, .gtf), etc...

Again, not to be used for data storage of large files!

Filesystems

/home/\$USER

/projects/\$USER

/scratch/alpine/\$USER

/pl/

Let's change directories into our scratch directory...



```
cd /scratch/alpine/$USER
```

Filesystems

```
/scratch/alpine/$USER
```

This is your largest directory, at 10TB!

This is the location where most of your data should be transferred into when you are ready for analysis

Any computation where the software or program is expected to write out output files, should be redirected to this directory space

This has the best read/write performance and compute nodes will be able to handle large volumes of data from this locations.

Filesystems

```
/scratch/alpine/$USER
```

This is your largest directory, at 10TB!

This is the location where most of your data should be transferred into when you are ready for analysis

Any computation where the software or program is expected to write out output files, should be redirected to this directory space

This has the best read/write performance and compute nodes will be able to handle large volumes of data from this locations.



However, there are a few caveats...

- Storage is temporary – files are automatically deleted 90 days from the date of creation/transfer
- Do NOT have your only copy of data here or you risk losing it after the allotted time

Filesystems

Use the following command to keep track of the creation date of each file and folder:

```
mmlsattr -L <myfile>
```



Filesystems

Use the following command to keep track of the creation date of each file and folder:

```
mmlsattr -L <myfile>
```

- With <myfile> being the name of the file or folder that you wish to track.
- Remember that with Linux, any folder or file is considered as a file!



Filesystems

/home/\$USER

/projects/\$USER

/scratch/alpine/\$USER

/pl/

For those of you that have a PetaLibrary allocation, you can access your allocation by using changing directories to the path that was given to you. Your allocation is mounted to your Alpine space automatically.



```
cd /pl/active/nameOfAllocation
```

Filesystems

/pl/

The benefits of having a petalibrary allocation is if you want permanent storage of data files so that you don't have to worry about constantly moving data in and out of your scratch space.

Consider a petalibrary allocation if:

- You have several TBs of data that you want permanently stored somewhere that Alpine has access to
- You don't want to manage moving files back and forth from scratch to your local computer

There is a cost associated with PL, which is ~\$45/TB/year, or \$65/TB/year if you want a second duplicate copy (recommended if you don't have a second copy save elsewhere) saved in case the hardware crashes

<https://www.colorado.edu/rc/resources/petalibrary>

Filesystems

/home/\$USER

2GB

- Do not use this for anything regarding data transfer, storage, software installs, etc...
- One access pre-existing configuration files here if you need to modify them (.bashrc, .bash_profile, .condrc, etc...)

/projects/\$USER

250GB

- Store all code, scripts, and sbatch files here
- Local software that you need to install yourself should go here
- If small enough, reference files that will be re-used, i.e. genome index, genome fasta or gtf annotation files, etc...

/scratch/alpine/\$USER

10TB

- Location for all data transfers and large data to be used to computation and analysis
- Output files generated by software should be redirected here



There is a 90-day automated data purge from the date of file creation/transfer; not to be used for permanent storage or any data that is single copy

/pl/

1TB-??

- OPTIONAL – only available to those that are paying for an allocation
- Permanent data storage mounted to your Alpine user space
- There is a cost associated with it

What is Petalibrary?

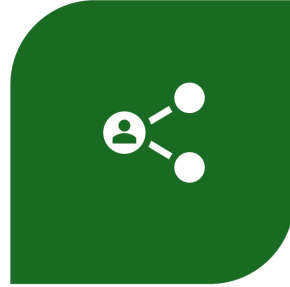
- An Alpine service that supports the paid storage, archival, and sharing of research data.
- It is available at a subsidized cost to any researcher affiliated with the University of Colorado System (Boulder, Anschutz, Denver, Colorado Springs)



Preliminary conditions



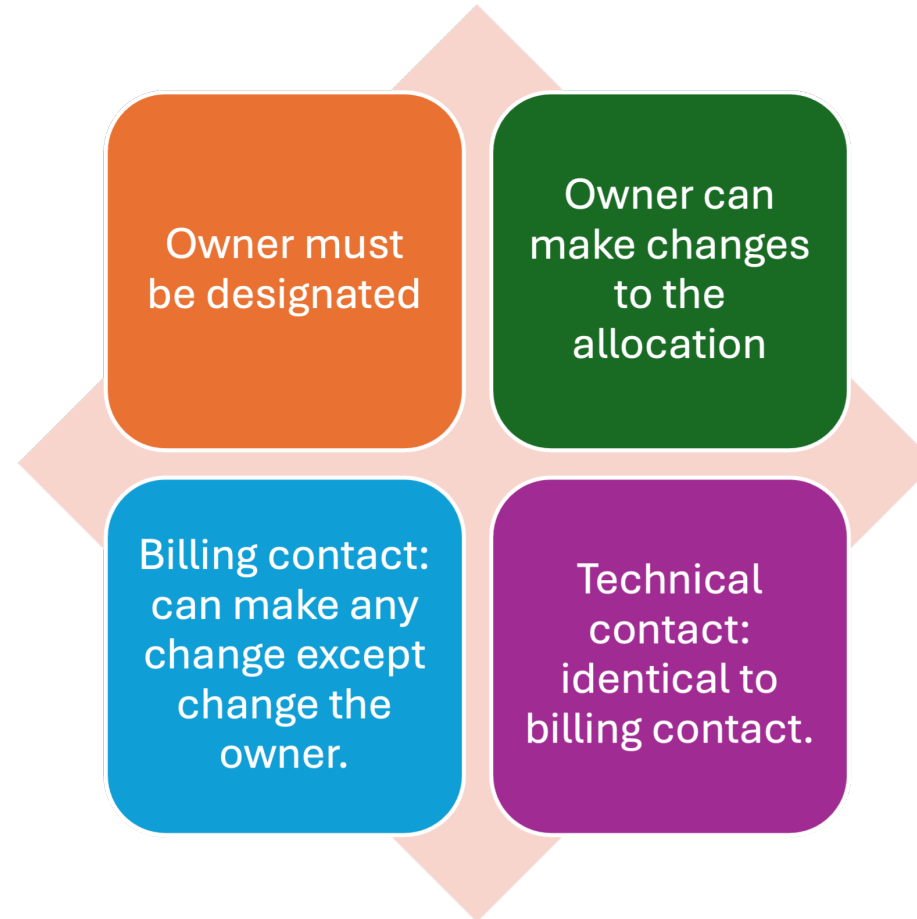
CREATION OF AN ACCESS
GROUP



ALL MEMBER ADDED TO THE
GROUP WILL NEED
ACCESS/XSEDE ACCOUNTS.

```
/pl/active/<your_allocation_name>  
/pl/archive/<your_allocation_name>
```

Preliminary conditions



Application

- Application submitted here:
<https://www.colorado.edu/rc/resources/petalibrary>
- The request form will need a speedtype: account# to which they plan to charge the allocation.



Billing

- On active storage: \$48/TB/yr.
- ZFS Raidz2 allow for frequent read/write + parity.
- **It is the responsibility of the PI to estimate and foresee the PL allotment for the year.**



Billing

- On archive storage: \$28/TB/yr.
- Tape-like storage for infrequently accessed data.
- Can be accessed from RC login node only and data transfer resources.
- Min size for any alloc is 1 TB.



Billing

- On archive storage: \$28/TB/yr.
- Tape-like storage for infrequently accessed data.
- Can be accessed from RC login node only and data transfer resources.
- Min size for any alloc is 1 TB.
- **10,000 object (file/directory) limit.**
- **Owner will get and pay for the space even if they do not use it.**



Data redundancy



PL allocations are of single-copy nature.



All users should fill out the PL single copy acknowledgment.



Snapshots monitoring in place so that they are not missed unless on snapshot custom schedule



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Data replication solution

- Replicated PetaLibrary active+archive allocation.
- CURC **only takes data replication responsibility.**



Data replication solution

- Replicated PetaLibrary active+archive allocation.
- CURC **only takes data replication responsibility.**
- **Not subject to the 10,000 object (file/directory) limit.**



Active+Archive replication

- Synchronization between active and archive happen every 15 min in theory but are not always guaranteed.
- Snapshots are also replicated from active to archive.
- The Boulder storage team maintains the second copy (e.g. archive) for you and you do not have direct access to it.
- \$76/TB/year total. **(NOT \$76 on top of the \$48/\$28 original rate!!)**



Archive DR

- CU Boulder storage team back it up monthly to an offsite location for you.
- More information here:
https://curc.readthedocs.io/en/latest/storage/petalibrary/allocation_types.html#archive-dr-disaster-recovery



Recap

Category	Feature	PetaLibrary Tier			
		Active	Archive	Active+Archive	Archive+DR
Rates per TB per year	CU System (FY27)	\$48	\$28	\$76	\$44
	External (FY27)	\$115	N/A	N/A	N/A
Data Integrity*	Snapshots	✓	✓	✓	✓
	Offsite replication			✓	
	Offsite backup				✓
	RAID double parity protection	✓		✓	
	RAID triple parity protection		✓		✓
	Full monthly data checksumming	✓	✓	✓	✓
Data Limits	No data volume limit	✓	✓	✓	✓
	Initial allocation request limit	200 T	200 T	200 T	200 T
	No object limit	✓		✓	
	10K object/TB limit		✓		✓
Data Access	Accessible from CURC compute nodes	✓		✓	
Data Transfer	Globus access	✓	✓	✓	✓
	rsync/scp/sftp/ access	✓	✓	✓	✓
	SMB access**	✓		✓	
Data Compression	lz4 data compression	✓		✓	
	zstd data compression		✓		✓

SMB mounts

- SMB mounts from the CU Anschutz campus to Alpine are not allowed due to safety concerns. Please refer to this page for more details:
https://curc.readthedocs.io/en/latest/petalibrary/allocation_types.html .

Petalibrary best practices.

- PL allocations fill up quickly due to snapshots if you use it on a daily basis.
- To preserve space, you may send an email to rc-help@colorado.edu to request the deletion of nonessential snapshots or a change of the snapshot scheduling.



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Alpine compute

Now you are on the head/login node. What are your options?

Option
A

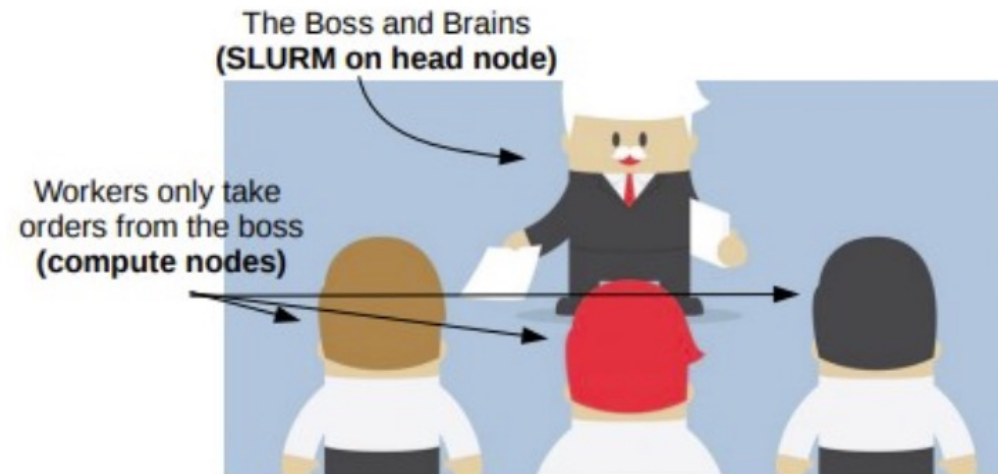
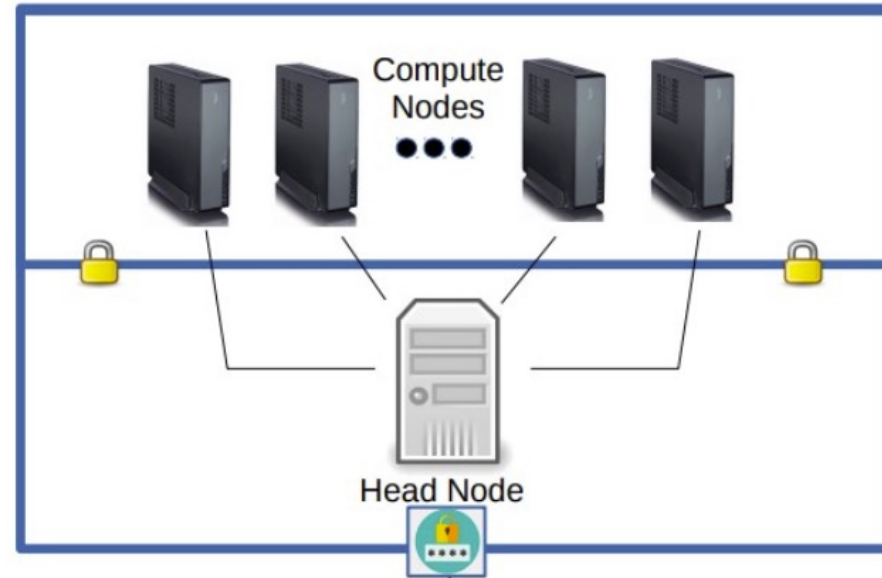
Make a request to the the head node to login to a compute node so you can interactively run your code on a compute node, test code in real-time, or look at what software is installed on Alpine.

Option
B

Stay on the head node because you want to submit a job remotely to a compute node.

Regardless of which option you choose, you still need to communicate that to the head/login node, so how do we do that?

- You can communicate via Simple Linux Utility for Resource Management (SLURM) job scheduling language
- The head/login node and compute nodes all speak SLURM, so your task is to translate what you want into SLURM format.
- SLURM is a workload manager and is just one of the many flavor of job scheduling softwares available for HPCs. You may have heard of LSF (bsub, bqueues, etc...) or PBS (qsub, qstat, etc...) which are other flavors that are commonly used in Academia and all accomplish the same task – scheduling jobs by orchestrating communication between head/login nodes and compute nodes.

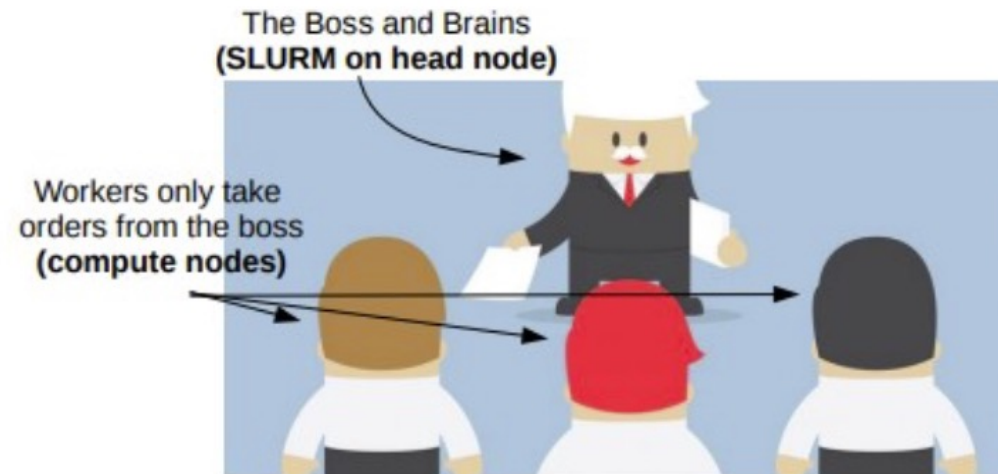
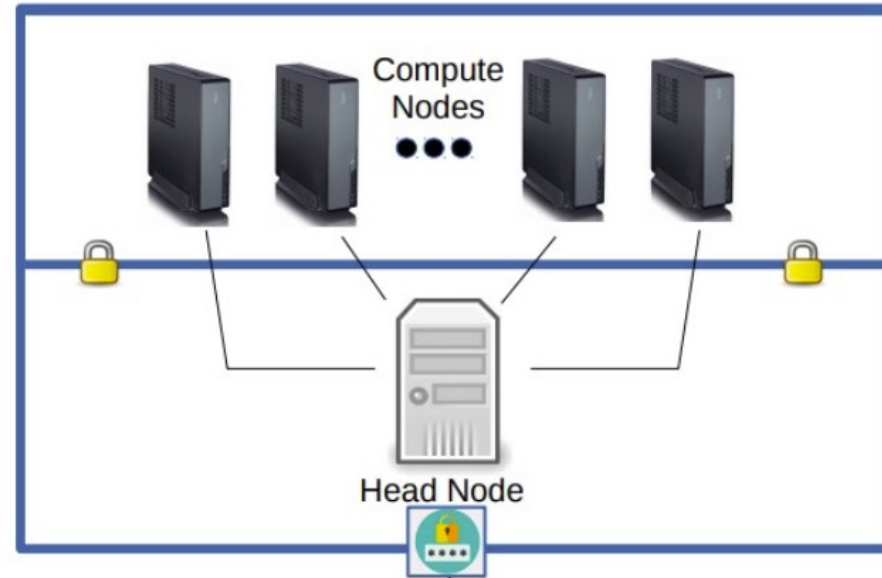


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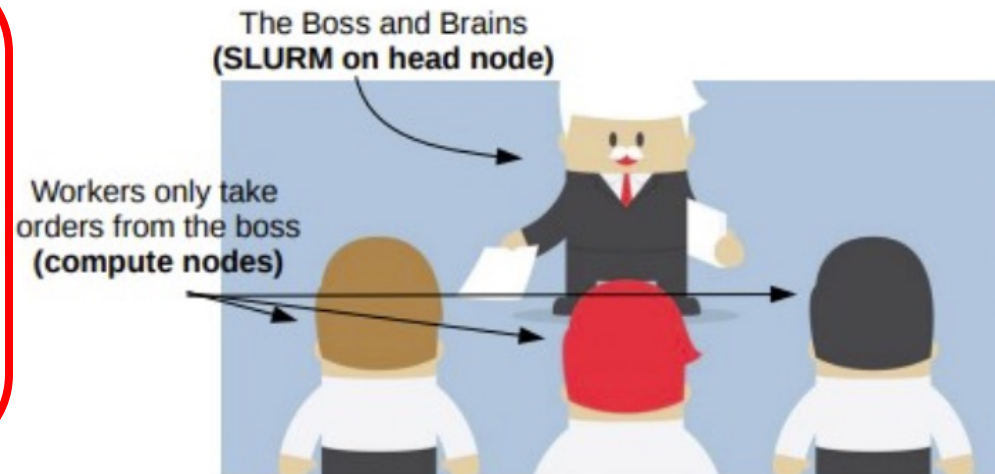
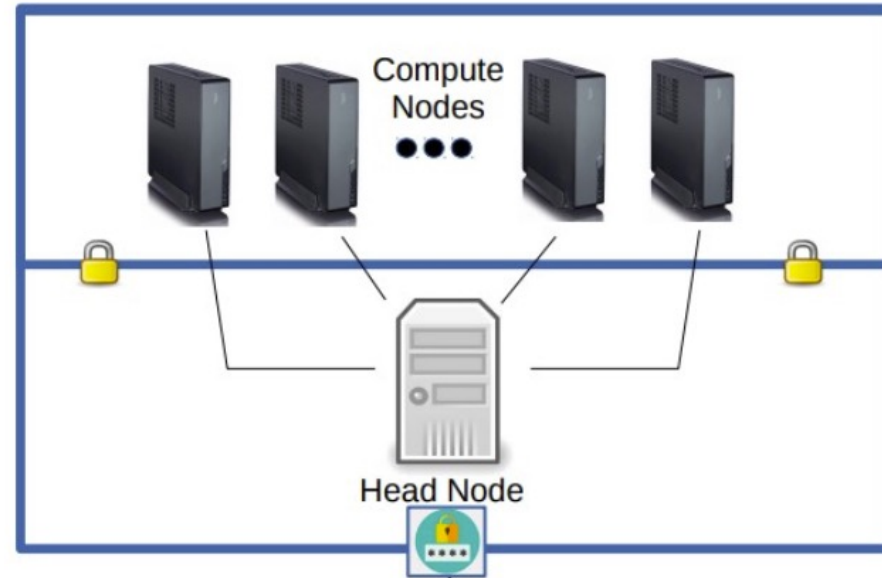
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HPCs use job schedulers

- What is a job scheduler?
 - Responsible for executing multiple job requests (i.e. your scripts and programs) whether they come from the same or different user(s) and determining what resources are available, and whose jobs are next in line to be run

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 - Responsible for executing multiple job requests (i.e. your scripts and programs) whether they come from the same or different user(s) and determining what resources are available, and whose jobs are next in line to be run
- Why do we need a scheduler?
 - “Fair-Share”
 - Sophisticated algorithm to schedule jobs fairly on a shared resource
 - The scheduler knows when each user made submitted requests so it can plan jobs effectively and know how long a job has been waiting in the queue or line
 - In the even all resources are being utilized, the “fair-share” policy is implemented meaning those users who have used the least amount of compute hours/resources within a window of time, get pushed to the front of the queue.
 - Prevents one individual from constantly using all the resources as the become available

HPCs use job schedulers

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 - In the even all resources are being utilized, the “fair-share” policy is implemented meaning those users who have used the least amount of compute hours/resources within a window of time, get pushed to the front of the queue.
 - Prevents one individual from constantly using all the resources as the become available
 - Efficiency
 - Jobs that are shorter or require less resources typically are in the queue shorter since resources become available more quickly and are more likely to fit into untapped resources
 - The scheduler know how many resources every user has requested and the time and memory constraints of those resources

Now you are on the head/login node. What are your options?

Option
A

Make a request to the the head node to login to a compute node so you can interactively run your code on a compute node, test code in real-time, or look at what software is installed on Alpine.

Option
B

Stay on the head node because you want to submit a job remotely to a compute node.

Option A: Request an interactive compute node

In case you get confused or are unsure if you are on the head/login node or a compute node you can check here! Anything that say login-ci followed by a number is the login/head node. I am on login-ci1, but you might have a different ci number

```
Host: login-ci1.rc.int.colorado.edu
Welcome to University of Colorado Boulder Research Computing!

Full documentation is available in our user guide at
https://www.rc.colorado.edu/support/user-guide. If you have a question
that's not answered there, contact us at rc-help@colorado.edu.

A number of directories have been created for you already:

* `/home/$USER`, your home directory
* `/projects/$USER`, your project directory

Run the command `module avail` to see a list of available software.

To prevent this README from being displayed at login, edit your
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* Questions? rc-help@colorado.edu
* Subscribe to system announcements: https://curc.statuspage.io/
* Please type rc-help for the Acceptable Use Policy and a short help page.

You are using login node: login-ci1

Users who had jobs in the queue prior to the planned maintenance should check
to confirm these jobs are still queued. Some jobs, particularly those schedule
since midnight today (Wed June 7), may have been canceled during the
maintenance period.
(base) [ ██████████@xsede.org login-ci1 ██████████@xsede.org ]$
```

Request an short
interactive compute
node

Now, we are inside of a compute node!

```
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You are using login node: login-cil

Users who had jobs in the queue prior to the planned maintenance should check
to confirm these jobs are still queued. Some jobs, particularly those schedule
since midnight today (Wed June 7), may have been canceled during the
maintenance period.
(base) [brunetti@xsede.org login-cil brunetti@xsede.org]$ acompile
acompile: submitting job... salloc --nodes=1 --partition=acompile --ntasks=1 --time=01:00:00 --qos=compile --job-name=acompile --bell --oversubscribe srun --pty /bin/bash
salloc: Granted job allocation 2650311
salloc: Nodes c3cpu-c15-u7-2 are ready for job
(base) [brunetti@xsede.org c3cpu-c15-u7-2 brunetti@xsede.org]$
```

You can now see that I am on a compute node. Some of the numbers and characters may be a bit different as there are 456 different compute nodes on Alpine as of July 2026 , but the big thing is you see it says `cpu` somewhere in there.

Now, we are inside of a compute node!

```
Welcome to University of Colorado Boulder Research Computing!

Full documentation is available in our user guide at
https://www.rc.colorado.edu/support/user-guide. If you have a question
that's not answered there, contact us at rc-help@colorado.edu.

A number of directories have been created for you already:
* `/home/$USER`, your home directory
* `/projects/$USER`, your project directory

Run the command `module avail` to see a list of available software.

To prevent this README from being displayed at login, edit your
`.bash_profile` or `.login` files.
Welcome to CU-Boulder Research Computing.

* Website http://colorado.edu/rc
* Questions? rc-help@colorado.edu
* Subscribe to system announcements: https://curc.statuspage.io/
* Please type rc-help for the Acceptable Use Policy and a short help page.

You are using login node: login-cl1

Users who had jobs in the queue prior to the planned maintenance should check
to confirm these jobs are still queued. Some jobs, particularly those schedule
since midnight today (Wed June 7), may have been canceled during the
maintenance period.

(base) [brunetti@xsede.org@login-cl1 brunetti@xsede.org]$ acompile
compile: submitting job... salloc --nodes=1 --partition=acompile --ntasks=1 --time=01:00:00 --qos=compile --job-name=acompile --bell --oversubscribe srun --pty /bin/bash
salloc: Granted job allocation 2650311
salloc: Nodes c3cpu-cl5-u7-2 are ready for job
(base) [brunetti@xsede.org@c3cpu-cl5-u7-2 brunetti@xsede.org]$
```

Notice that if you call `acompile` without any specifications it shows these parameters. This means you are requesting a CPU with 1 core (`--ntasks=1`), for 1 hour (`--time=01:00:00`), which is the default



This means that your session will automatically close/get killed if you use it longer than 1 hour.



The command below from the head node would not kill your job until 12 hours, which is also the maximum time you can request and interactive job

```
[brunetti@xsede.org@login-cl1 brunetti@xsede.org]$ acompile --time=12:00:00
```

Option A: Things we can do once we are interactively inside of a compute node...

- Check what software is installed on Alpine
- Install our own software
- Run software and code interactively

LMOD (CONTEXT)

- Modern environment module system for HPC
- Initially introduced by Robert McLay from TACC (UT Austin) in 2011.



LMOD (CONTEXT)

- Users have on HPC have different needs.
- Applications, compilers, libraries, versions being used might be different.



LMOD (benefits)

- Users do not need to know where software is installed
- Environment variable to interface packages can be set (e.g. picard).
- Very useful for software reproducibility



LMOD (benefits)

- Supports software hierarchy
- Users can set their environment modules at will.



Option A: Things we can do once we are interactively inside of a compute node...

- Check what software is installed on Alpine



```
module avail
```

Option A: Things we can do once we are interactively inside of a compute node...

- Check what software is installed on Alpine



module avail

```
base) [redacted]@xsede.org[c3cpu-c11-u17-2 [redacted]xsede.org]$ module avail

----- /curc/sw/modules/slurm -----
StdEnv  curc-quota/latest (D)  slurm/alpine (D)  slurm/blanca  slurm/core  slurmtools (D)
----- /usr/share/lmod/lmod/modulefiles/Core -----
lmod  settarg

----- Compilers -----
aocc/3.1.0 (D)  aocc/3.2.0  gcc/10.3.0  gcc/11.2.0 (D)  intel/2022.1.2 (M)  nvhpc_sdk/2022.229  nvhpc_sdk/2023.233 (D)

----- Independent Applications -----
R/3.6.3  cuda/11.3 (g)  ghostscript/9.56.0  julia/1.6.6  paraview/5.0.1  rclone/1.58.0  tdom/0.9.2 (D)
R/4.2.2 (D)  cuda/11.4 (g)  git-lfs/3.1.2  julia/1.8.1 (D)  paraview/5.6.0  rhel7for8/1.0  texlive/2021
allinea/6.0.4 (M)  cuda/11.8 (g)  git/2.31.0  lftp/4.8.4  paraview/5.9.0  rocm/5.2.3 (g)  totalview/2016.
6.21
anaconda/2020.11  cuda/12.1.1 (g,D)  gms/2.16.0  loadbalance/0.2  paraview/5.10.0 (D)  rocm/5.3.0 (g)  ucx/1.10.1 (D)
anaconda/2022.10 (D)  cudnn/8.1 (g)  gms/4.11.1 (D)  mambaforge/23.1.0-1  pdttoolkit/3.22  rocm/5.5.0 (g,D)  ucx/1.12.1 (D)
arm-forge/19.1.3 (M)  cudnn/8.2 (g)  gnu_parallel/20160622  mathematica/9.0  perl/5.16.3  ruby/2.3.1  udunits/2.2.20
autotools/2.69  cudnn/8.6 (g,D)  gnu_parallel/20210322 (D)  mathematica/11.1.0 (D)  perl/5.24.0 (D)  ruby/3.0.0 (D)  udunits/2.2.24
autotools/2.71 (D)  curc-quota/latest  gnuplot/5.4.3  matlab/R2018b  perl/5.28.1 (D)  singularity/3.6.4 (D)  udunits/2.2.25
chimerax/1.2.5  dmtcp/2.6.0  idl/8.7  matlab/R2019b  perl/5.36.0  singularity/3.7.4  udunits/2.2.28 (D)
cmake/3.5.2  eigen/3.4.0  imagenagick/6.9.12  matlab/R2020b  perl/5.36.0  slurmtools/0.0.0  valgrind/3.11.0
cmake/3.9.2  emacs/25.3  jdk/1.7.0  matlab/R2021b (D)  pigz/2.7  slurmtools/0.0.1  valgrind/3.17.0 (D)
cmake/3.14.1  emacs/27.2 (D)  jdk/1.8.0_91  matlab/R2022b  python/2.7.18  subversion/1.8.16  vapor/3.3.0
cmake/3.20.2  expat/2.1.1 (D)  jdk/1.8.0_281  maven/3.8.1  python/3.10.2 (D)  subversion/1.10.2  vapor/3.4.0 (D)
cmake/3.25.0 (D)  expat/2.3.0 (D)  jdk/1.8.0  ncl/6.3.0  qchem/4010  subversion/1.14.1 (D)  vtf3/1.43
cube/3.4.3  ffmpeg/4.4  jdk/18.0.1.1 (D)  ncl/6.6.2 (D)  qt/5.6.0  tccltk/8.6.5  zip/rhel7
cube/4.3.4 (D)  gdb/8.1  julia/0.6.2  papi/5.4.3  qt/5.9.1  tccltk/8.6.11 (D)
cuda/11.2 (g)  gdb/10.1 (D)  julia/1.6.0  papi/5.5.1 (D)  qt/5.15  tdom/0.8.3

----- Bioinformatics -----
alphafold/2.2.0  bbtools/39.01  bowtie2/2.5.0  cutadapt/4.2  homer/4.11  nextflow/22.10.6  plink2/2.00a2.3  sra-toolkit/3.0.0
alphafold/2.3.1 (D)  bcftools/1.16  bwa/0.7.17  fastqc/0.11.9  htstlb/1.16  nextflow/23.04 (D)  qlime2/2023.5  star/2.7.10b
bamtools/2.5.2  bedtools/2.29.1  cellranger/7.1.0  gatk/4.3.0.0  multiqc/1.14  picard/2.27.5  samtools/1.16.1  trimmomatic/0.39

----- Lmod Internal Modules -----
StdEnv  lmod  settarg

Where:
g: built for GPU
M: built for host and native MIC
D: Default Module

Use "module spider" to find all possible modules and extensions.
Use "module keyword key1 key2 ..." to search for all possible modules matching any of the "keys".
```

Option A: Things we can do once we are interactively inside of a compute node...

- Check what software is installed on Alpine



module avail



Just a reminder! **I have to be inside of a compute node** to see this, not the head/login node!!

```
base) [redacted@redacted xse.org(c3cpu-c11-u17-2 [redacted]@xse.org)]$ module avail

----- /curc/sw/modules/slurm -----
StdEnv  curc-quota/latest (D)  slurm/alpine (D)  slurm/blanca  slurm/core  slurmtools (D)
----- /usr/share/lmod/lmod/modulefiles/Core -----
lmod  settarg

----- Compilers -----
aocc/3.1.0 (D)  aocc/3.2.0  gcc/10.3.0  gcc/11.2.0 (D)  intel/2022.1.2 (m)  nvhpc_sdk/2022.229  nvhpc_sdk/2023.233 (D)

----- Independent Applications -----
R/3.6.3  R/4.2.2  allinea/6.0.4 (m)  anaconda/2020.11  anaconda/2022.10 (D)  arm-forge/19.1.3 (m)  autotools/2.69  autotools/2.71 (D)  chimeraX/1.2.5  cmake/3.5.2  cmake/3.9.2  cmake/3.14.1  cmake/3.20.2  cmake/3.25.0 (D)  cube/3.4.3  cube/4.3.4 (D)  cuda/11.2  cuda/11.3  cuda/11.4  cuda/11.8  cudnn/8.1  cudnn/8.2  cudnn/8.6  curc-quota/latest  dmtcp/2.6.0  eigen/3.4.0  emacs/25.3  emacs/27.2  expat/2.1.1  expat/2.3.0  ffmpeg/4.4  gdb/8.1  gdb/10.1  ghostscript/9.56.0  glib-lfs/3.1.2  git/2.31.0  gms/2.16.0  gms/4.11.1  gnu_parallel/20160622  gnu_parallel/20210322 (D)  gnuplot/5.4.3  idl/8.7  imagemagick/6.9.12  jdk/1.7.0  jdk/1.8.0_91  jdk/1.8.0_281  jdk/1.8.0  jdk/18.0.1.1  julia/0.6.2  julia/1.6.0  julia/1.6.6  julia/1.8.1 (D)  lftp/4.8.4  loadbalance/0.2  mambaforge/23.1.0-1  mathematica/9.0  mathematica/11.1.0 (D)  matlab/R2018b  matlab/R2019b  matlab/R2020b  matlab/R2021b  matlab/R2022b  maven/3.8.1  ncl/6.3.0  ncl/6.6.2  papi/5.4.3  papi/5.5.1  paraview/5.0.1  paraview/5.6.0  paraview/5.9.0  pdttoolkit/3.22  pdttoolkit/3.25.1 (D)  perl/5.16.3  perl/5.24.0  perl/5.28.1  perl/5.36.0  pigz/2.7  python/2.7.18  python/3.10.2 (D)  qchem/4010  qt/5.6.0  qt/5.9.1  qt/5.15  rclone/1.58.0  rhel7for8/1.0  rocm/5.2.3  rocm/5.3.0  rocm/5.5.0  ruby/2.3.1  ruby/3.0.0  singularity/3.6.4 (D)  singularity/3.7.4  slurmtools/0.0.0  slurmtools/0.0.1  subversion/1.8.16  subversion/1.10.2  subversion/1.14.1 (D)  tcltk/8.6.5  tcltk/8.6.11 (D)  tdom/0.9.2  texlive/2021  totalview/2016.  ucx/1.10.1  ucx/1.12.1  udunits/2.2.20  udunits/2.2.24  udunits/2.2.25  udunits/2.2.28  valgrind/3.11.0  valgrind/3.17.0  vapor/3.3.0  vapor/3.4.0  vtf3/1.43  zip/rhel7

----- Bioinformatics -----
alphafold/2.2.0  alphafold/2.3.1 (D)  bamtools/2.5.2  bbtools/39.01  bcftools/1.16  bedtools/2.29.1  bowtie2/2.5.0  bwa/0.7.17  cellranger/7.1.0  cutadapt/4.2  fastqc/0.11.9  gatk/4.3.0.0  homer/4.11  htlib/1.16  multiqc/1.14  nextflow/22.10.6  nextflow/23.04 (D)  picard/2.27.5  plink2/2.00a2.3  qtime2/2023.5  samtools/1.16.1  sra-toolkit/3.0.0  star/2.7.10b  trimomatic/0.39

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D: Default Module

Use "module spider" to find all possible modules and extensions.
Use "module keyword key1 key2 ..." to search for all possible modules matching any of the "keys".
```

Alpine software

- Refer to this list to look for all the already installed software on Alpine:
<https://curc.readthedocs.io/en/latest/clusters/alpine/software.html>
- We will cover this topic more in depth during the workshop: Alpine module stack and miniforge.



Option A: Things we can do once we are interactively inside of a compute node...

- Check what software is installed on Alpine

- Install our own software

- Run software and code interactively

Option A: Things we can do once we are interactively inside of a compute node...

- **Install our own software**

You will notice that not all software is installed on Alpine but you can install Alpine yourself locally. There are typically two recommended ways of doing this:

- Through a [miniforge \(preferred\)](#) or anaconda environment
- Following the website instructions to install from source locally

Option A: Things we can do once we are interactively inside of a compute node...

- **Install our own software – using** miniforge or anaconda



Recall, that your home directory is really, really small (only 2GB), so we always want to make sure software is installed in our projects directory (`/projects/$USER/`). By default conda likes to install in home, so before do any conda work, we must change our `.condarc` file to include the following lines:

```
pkgs_dirs:  
  - /projects/$USER/.conda_pkgs  
envs_dirs:  
  - /projects/$USER/software/anaconda/envs
```

This only every needs to be done 1 time for the lifetime of your Alpine account!

Option A: Things we can do once we are interactively inside of a compute node...

- **Through a** miniforge or anaconda environment

1 Navigate to your Home Directory within your OnDemand session:

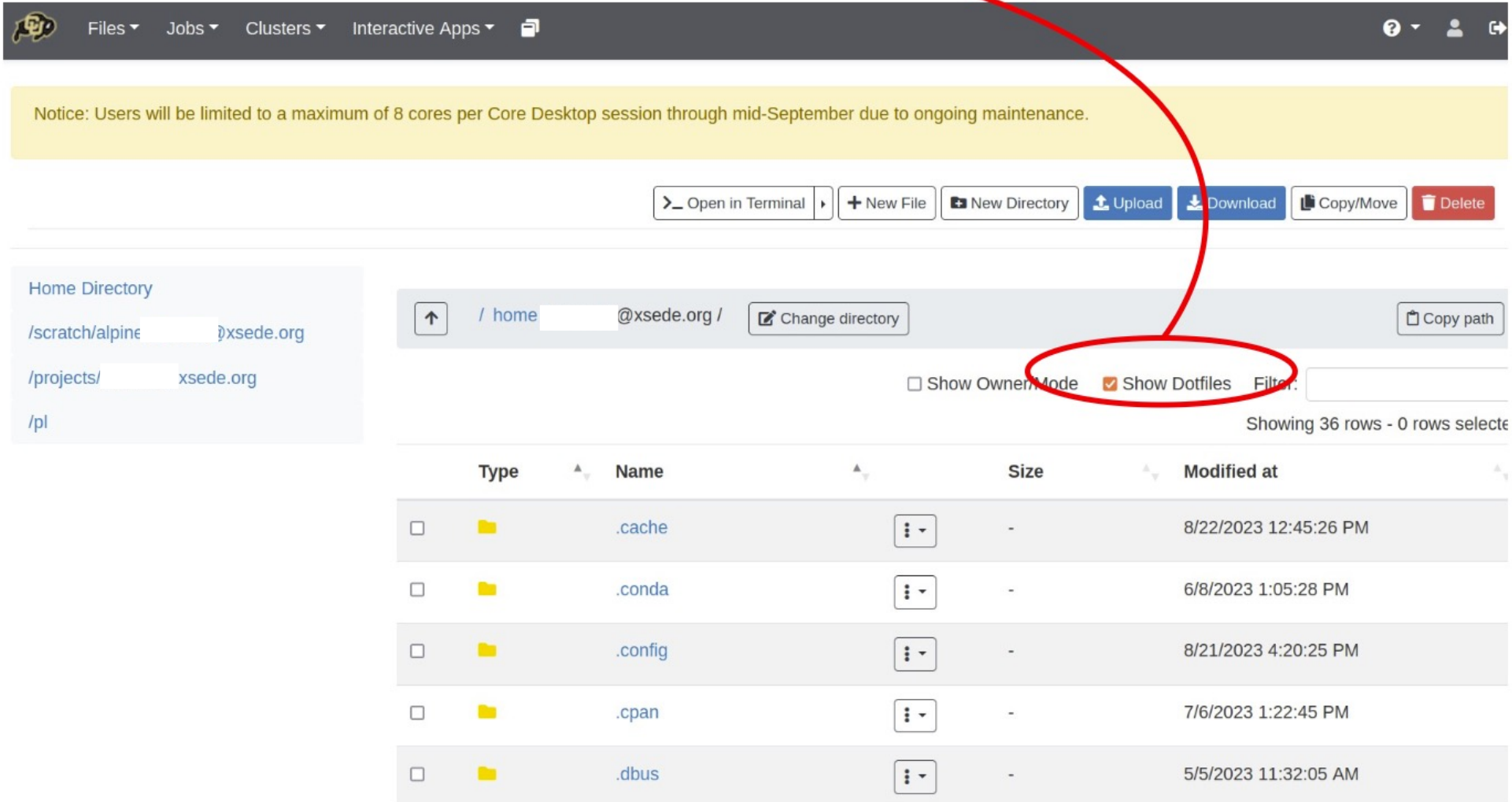
The screenshot shows the OnDemand web interface. At the top, a dark navigation bar contains links for Files, Jobs, Clusters, and Interactive Apps. A red circle highlights the 'Files' dropdown menu, which is open and shows the 'Home Directory' option selected. Below the navigation bar, a yellow banner displays a notice: 'Notice: Resources per Core Desktop session through mid-September due to ongoing maintenance.' The main content area is divided into two sections. On the left, a sidebar titled 'Home Directory' lists the following paths: '/scratch/alpine/ [username]@xsede.org', '/projects [username]@xsede.org', and '/pl'. On the right, the main file listing area shows the current directory as '/ home / [username]@xsede.org /'. Above the file listing are buttons for 'Open in Terminal', 'New File', 'New Directory', 'Upload', 'Download', 'Copy/Move', and 'Delete'. Below the directory path are checkboxes for 'Show Owner/Mode' and 'Show Dotfiles', and a 'Filter:' input field. The file listing table shows 3 of 36 rows, with 0 rows selected. The table has columns for Type, Name, Size, and Modified at. The files listed are:

Type	Name	Size	Modified at
Folder	ondemand	-	9/6/2022 3:39:58 PM
Folder	perl5	-	7/6/2023 1:20:16 PM
File	README.mdwn	562 Bytes	2/1/2018 8:35:24 AM

Option A: Things we can do once we are interactively inside of a compute node...

- **Through a miniforge or anaconda environment**

2 Make sure to click on the box that say “Show Dotfiles”



The screenshot shows the XSEDE file manager interface. A red circle highlights the 'Show Dotfiles' checkbox, which is checked. A red arrow points from the text 'Make sure to click on the box that say “Show Dotfiles”' to this checkbox. The interface includes a top navigation bar with links for Files, Jobs, Clusters, and Interactive Apps. A yellow notice banner states: 'Notice: Users will be limited to a maximum of 8 cores per Core Desktop session through mid-September due to ongoing maintenance.' Below the navigation bar are buttons for 'Open in Terminal', 'New File', 'New Directory', 'Upload', 'Download', 'Copy/Move', and 'Delete'. The main area shows the current directory as '/home/[username]@xsede.org /'. On the left, there is a 'Home Directory' sidebar with links to '/scratch/alpine/[username]@xsede.org', '/projects/[username]@xsede.org', and '/pl'. The file list displays several hidden files (dotfiles) including .cache, .conda, .config, .cpan, and .dbus. The table has columns for Type, Name, Size, and Modified at.

Type	Name	Size	Modified at
Folder	.cache	-	8/22/2023 12:45:26 PM
Folder	.conda	-	6/8/2023 1:05:28 PM
Folder	.config	-	8/21/2023 4:20:25 PM
Folder	.cpan	-	7/6/2023 1:22:45 PM
Folder	.dbus	-	5/5/2023 11:32:05 AM

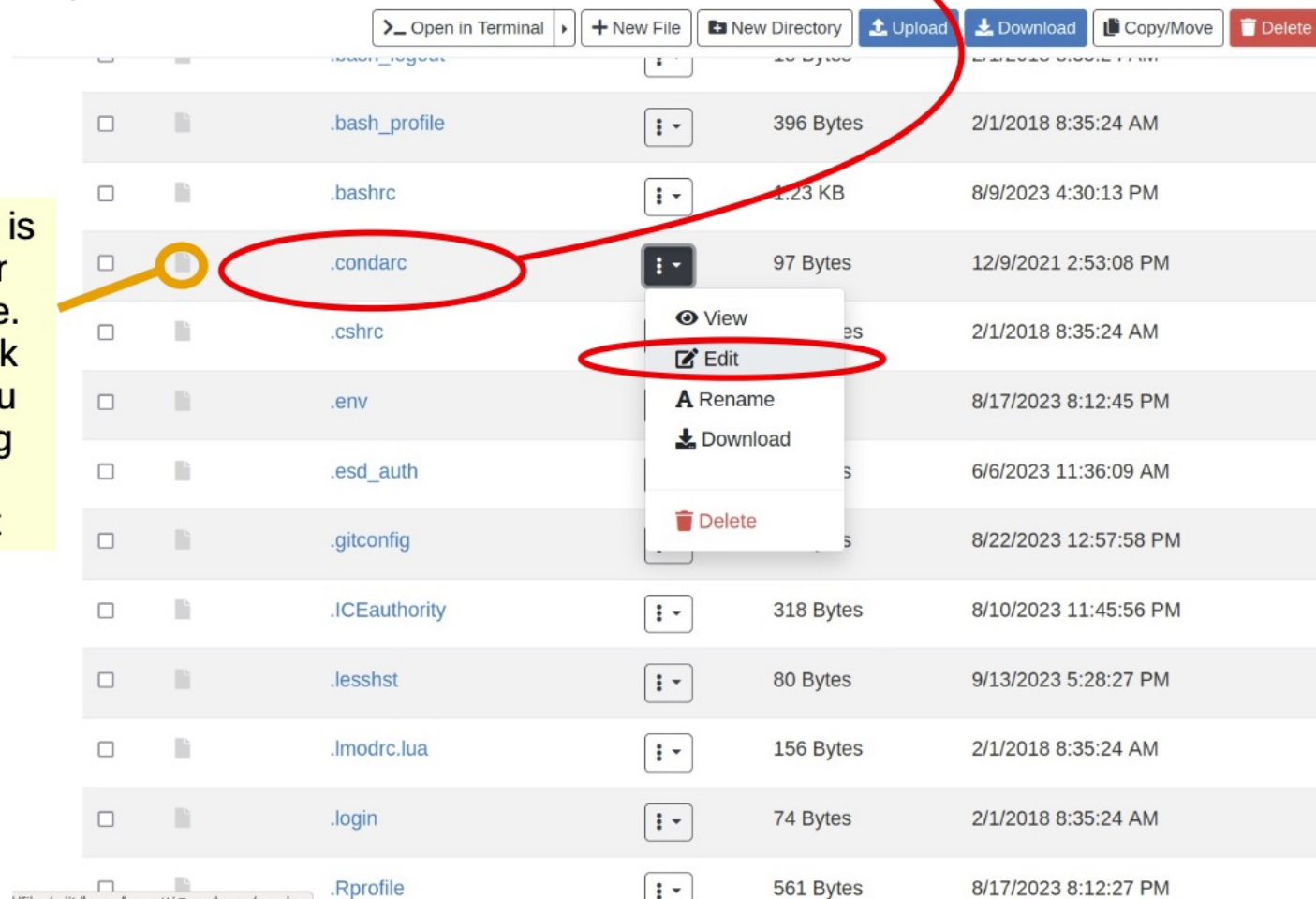
Option A: Things we can do once we are interactively inside of a compute node...

- **Through a** miniforge or anaconda environment

3 Scrolls down your list and find the file called `.condarc` and click the three dots and select Edit from the drop down menu



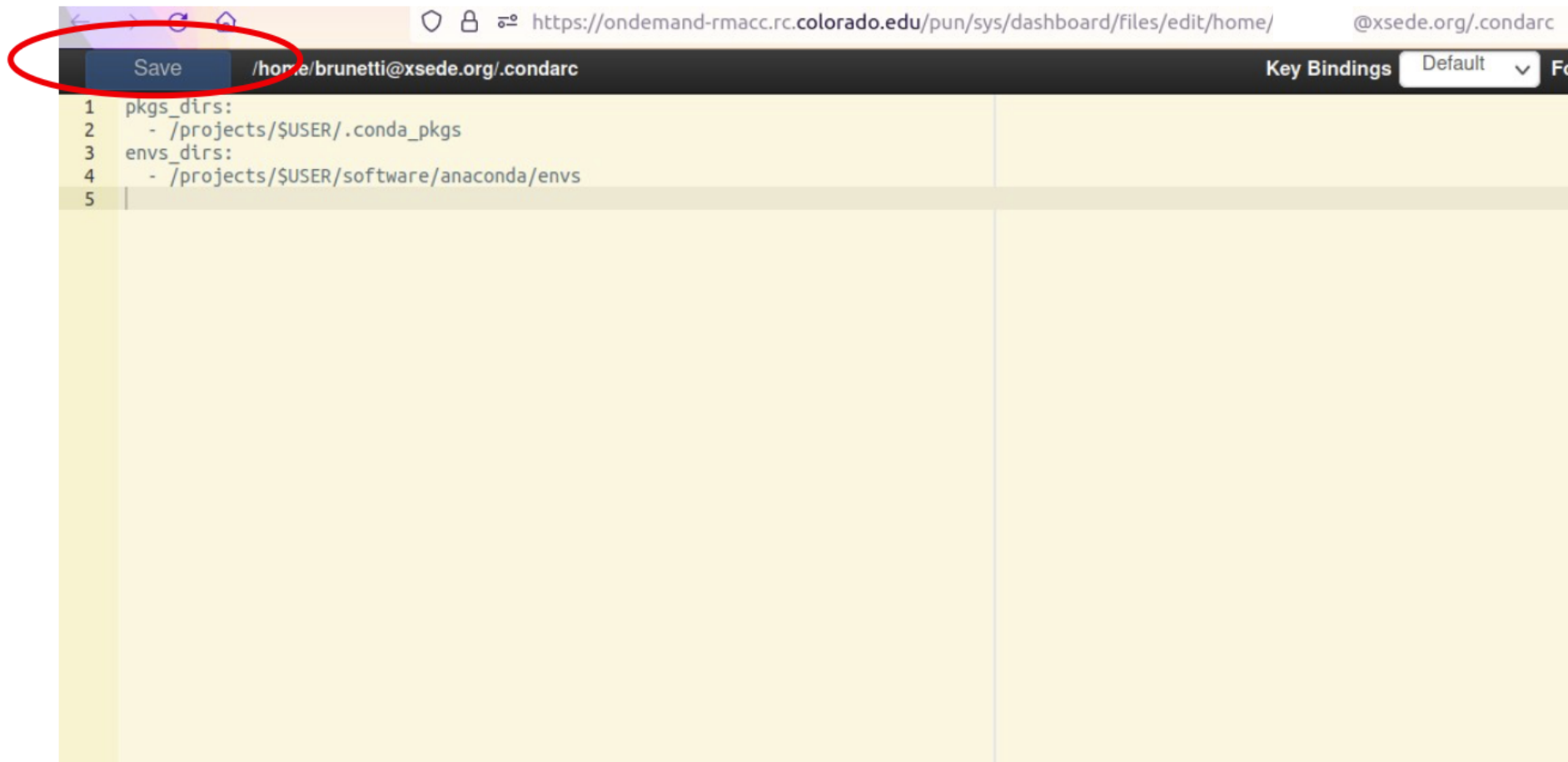
Be careful! There is a `.condarc` folder and a `.condarc` file. Make sure to click the file, which you can tell by seeing the little paper image to the left



Option A: Things we can do once we are interactively inside of a compute node...

- **Through a** miniforge or anaconda environment

4 This takes you to a text editor where you can copy and paste the following lines. Then be sure to press Save in the upper left hand corner



The screenshot shows a web browser window with the address bar displaying `https://ondemand-rmacc.rc.colorado.edu/pun/sys/dashboard/files/edit/home/` and a user identifier `@xsede.org/.condarc`. The editor's title bar shows a `Save` button and the file path `/home/brunetti@xsede.org/.condarc`. The main editing area contains the following text:

```
1 pkgs_dirs:  
2   - /projects/$USER/.conda_pkgs  
3 envs_dirs:  
4   - /projects/$USER/software/anaconda/envs  
5
```

Option A: Things we can do once we are interactively inside of a compute node...

- **Through a** miniforge or anaconda environment

5 After you press save, you can exit the browser tab and you are done configuring conda!

Option A: Things we can do once we are interactively inside of a compute node...

- **Through a** miniforge or anaconda environment

5 After you press save, you can exit the browser tab and you are done configuring conda!

Now that conda is configured, let's go through a tutorial of how to install software that has a conda installation option.

Let's try to install the commonly used bioinformatics package, RSEM, which is a way to count/quantify reads mapping to a genome.

Option A: Things we can do once we are interactively inside of a compute node...

- **Through a** miniforge or anaconda environment

5 After you press save, you can exit the browser tab and you are done configuring conda!

Now that conda is configured, let's go through a tutorial of how to install software that has a conda installation option.

Let's try to install the commonly used bioinformatics package, RSEM, which is a way to count/quantify reads mapping to a genome.

Please refer to the miniforge guide for more details:
https://github.com/kf-cuanschutz/CU-Anschutz-HPC-documentation/blob/main/miniforge_migration.md

Option A: Things we can do once we are interactively inside of a compute node...

- Check what software is installed on Alpine
- Install our own software
- Run software and code interactively

Option A: Things we can do once we are interactively inside of a compute node...

- **Run software and code interactively**

Once you are in a compute node you can also run code interactively, similar to how you have been doing on our command line. You may have noticed that Alpine also has several programming languages already installed that can be run on the command line and any other software (STAR, cutadapt, fastqc, Python, R, Ruby, Perl, C/C++, Java, Matlab, Mathematica, Julia, etc...)



----- /curc/sw/modules/slurm -----					
StdEnv (L,D)	curc-quota/latest (L,D)	slurm/alpine (L,D)	slurm/blanca	slurm/core	slurmtools (D)
----- /usr/share/lmod/lmod/modulefiles/Core -----					
StdEnv	lmod (D)	settag (D)			
----- Compilers -----					
aocc/3.1.0 (D)	gcc/11.2.0	intel/2022.1.2 (m)	nvhpc_sdk/2022.229	nvhpc_sdk/2025.255	
aocc/3.2.0	gcc/13.2.0	intel/2024.2.1 (D)	nvhpc_sdk/2023.233 (D)		
gcc/10.3.0	gcc/14.2.0 (D)	nvhpc_sdk/2021.213	nvhpc_sdk/2025.251		
----- Independent Applications -----					
R/3.6.3	emacs/27.2	mathematica/11.1.0 (D)	rocm/5.5.0	(g)	
R/4.2.2	emacs/30.1 (D)	matlab/R2018b	rocm/5.6.0	(g)	
R/4.4.0 (D)	expat/2.1.1	matlab/R2019b	rocm/6.1.0	(g,D)	
allinea/6.0.4 (m)	expat/2.3.0 (D)	matlab/R2020b	ruby/2.3.1		
anaconda/2020.11	ffmpeg/4.4	matlab/R2021b (D)	ruby/3.0.0	(D)	
anaconda/2022.10	gdb/8.1	matlab/R2022b	singularity/3.6.4	(D)	
anaconda/2023.09 (D)	gdb/10.1 (D)	matlab/R2023b	singularity/3.7.4		
antlr/4.13.1	ghostscript/9.56.0	matlab/R2024b	slurmtools/0.0.1		
arm-forge/19.1.3 (m)	git-lfs/3.1.2	maven/3.8.1	spack/0.20.1		
autotools/2.69	git/2.31.0	miniforge/24.11.3-0	subversion/1.8.16		
autotools/2.71 (D)	gmsh/2.16.0	ncl/6.3.0	subversion/1.10.2		
chimerax/1.2.5	gmsh/4.11.1 (D)	nco/4.8.1	subversion/1.14.1	(D)	
clustershell/1.9.2	gnu_parallel/20160622	papi/5.4.3	swig/4.1.1		
cmake/3.5.2	gnu_parallel/20210322 (D)	papi/5.5.1 (D)	tcltk/8.6.5		
cmake/3.9.2	gnuplot/5.4.3	paraview/5.0.1	tcltk/8.6.11		
cmake/3.14.1	idl/8.7	paraview/5.6.0	tcltk/9.0.1	(D)	
cmake/3.20.2	imagemagick/6.9.12	paraview/5.8.0	tdom/0.8.3		
cmake/3.25.0	jdk/1.7.0	paraview/5.9.0	tdom/0.9.2	(D)	
cmake/3.27.7	jdk/1.8.0_91	paraview/5.10.0 (D)	texlive/2021		

--More--

Option A: Things we can do once we are interactively inside of a compute node...

- **Run software and code interactively**

Let's say I wanted to test out some code in R in real-time. I can go inside of a compute node, load the programming module and start it by typing in the name of the programming language/software.



```
module load R/ 4.4.0
```

Followed by entering:



```
R
```

```
R version 4.2.2 (2022-10-31) -- "Innocent and Trusting"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> 
```

Now we are in an R session! We can install libraries, load libraries, write code and it will run; I am not going to talk about this much more in that this is really only good for testing since if you close your computer, turn it off etc... it will kill your session.

Option A: Things we can do once we are interactively inside of a compute node...

- **Run software and code interactively**

Let's say I wanted to test out some code in R in real-time. I can go inside of a compute node, load the programming module and start it by typing in the name of the programming language/software.



```
module load R/ 4.4.0
```



Do not use
an R
version
lower than
4.4.0!

Followed by entering:



```
R
```

```
R version 4.2.2 (2022-10-31) -- "Innocent and Trusting"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> 
```

Now we are in an R session! We can install libraries, load libraries, write code and it will run; I am not going to talk about this much more in that this is really only good for testing since if you close your computer, turn it off etc... it will kill your session.

R on Alpine

- We also have an existing guide here: https://github.com/kf-cuanschutz/CU-Anschutz-HPC-documentation/blob/main/Workshops/R_guide_on_Alpine_v2.pdf
- Rstudio package installation is also covered here: https://github.com/kf-cuanschutz/CU-Anschutz-HPC-documentation/tree/main/Rstudio_related_scripts

Best practices on Alpine

- /home or /tmp have a very small size. Please always include the following in your \$HOME/.bashrc or Slurm script.

```
export TMP=/gpfs/alpine1/scratch/$USER/cache_dir
```

```
mkdir -pv $TMP
```

```
export TEMP=$TMP
```

```
export TMPDIR=$TMP
```

```
export TEMPDIR=$TMP
```

```
export PIP_CACHE_DIR=$TMP
```



Best practices on Alpine

s ▾ Interactive Apps ▾ 

 ▾  

>_ Open in Terminal ▸

 Refresh

 New File

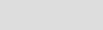
 New Directory

 Upload

 Download

 Copy/Move

 Delete

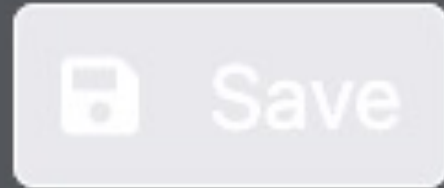
 / home /  @xsede.org /  Change directory  Copy path

☐ Show Owner/Mode ☒ Show Dotfiles Filter:

Showing 5 of 415 rows - 0 rows selected

☐	Type	▲	Name		Size	Modified at
☐			.bashrc		4.68 kB	5/13/2025 11:05:24 AM

Best practices on Alpine



/home/[REDACTED]@xsede.org/.bashrc

```
124 export TMP=/gpfs/alpine1/scratch/$USER/cache_dir
125 mkdir -pv $TMP
126 export TEMP=$TMP
127 export TMPDIR=$TMP
128 export TEMPDIR=$TMP
129 export PIP_CACHE_DIR=$TMP
```



Best practices on Alpine

- In general, always make sure to check your .cache in \$HOME to make sure you are not running out of space.
- To clean up your .cache space, using the shell please run the following:

```
rm -Rv $HOME/.cache/*
```



Username aliasing

- Some packages that use perl won't install or run Alpine due to the @xsede.org username format.
- Please follow this guide if that happens to you:
<https://curc.readthedocs.io/en/latest/additional-resources/csu-xsede-usernames.html>



Now you are on the head/login node. What are your options?

Option
A

Make a request to the the head node to login to a compute node so you can interactively run your code on a compute node, test code in real-time, or look at what software is installed on Alpine.

Option
B

Stay on the head node because you want to submit a job remotely to a compute node.

Elements of computing

Parts of the processing chip



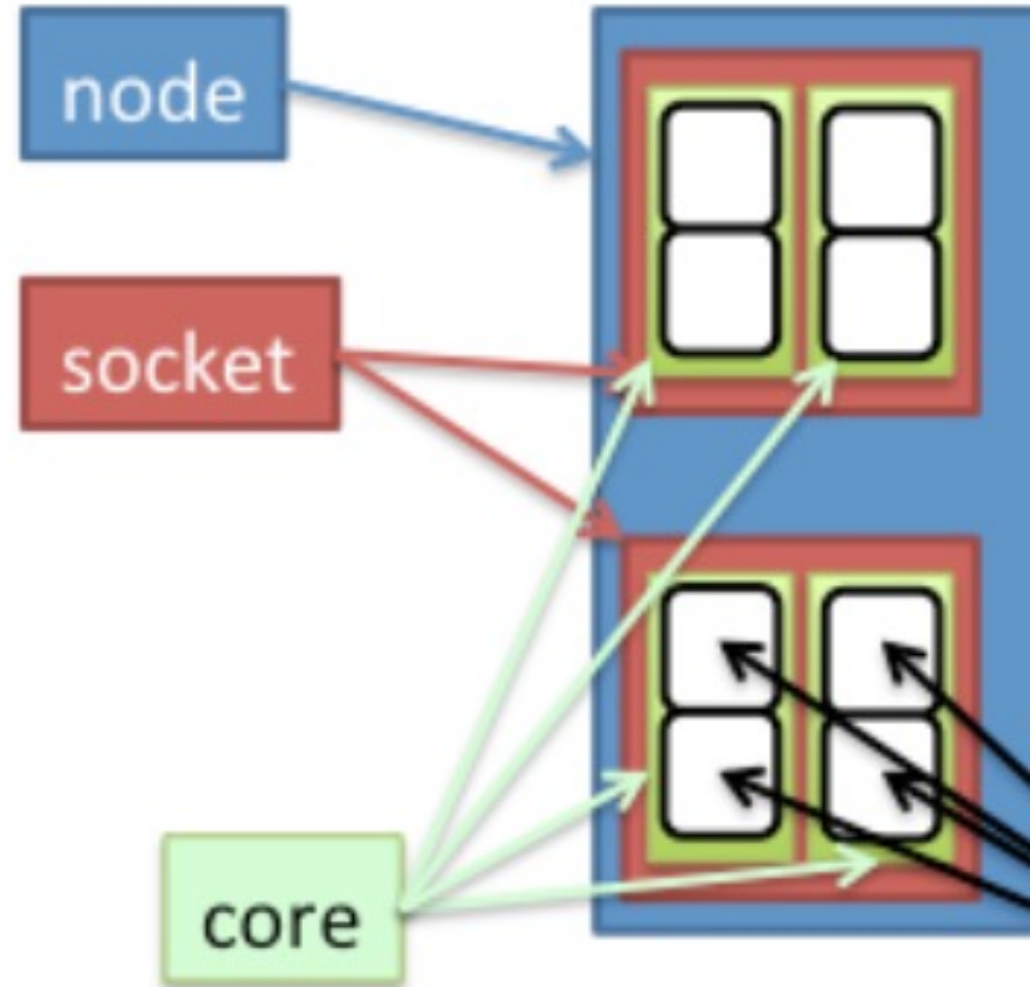
- A socket can be considered an entire computing chip in modern computers.
- Contains more than 1 core.
- On supercomputers a node usually contains more than 1 socket.



Elements of computing

A core contains an Arithmetic and Logic Unit (**ALU**) + Registers

Parts of the processing chip



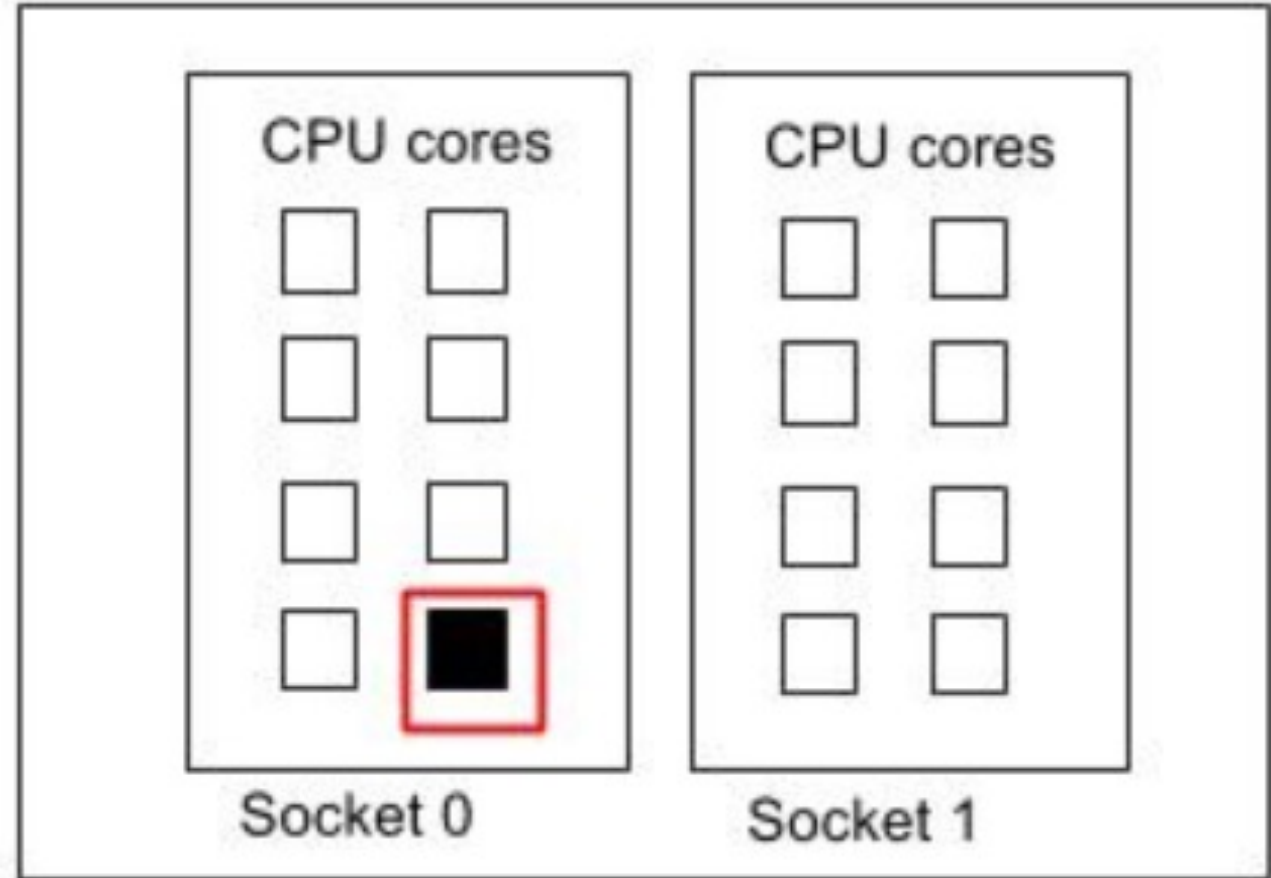
Source: <https://hpc.nih.gov/apps/swarm.html>



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Elements of computing

**Parts of the
processing chip**



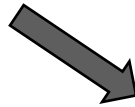
Source: <https://hpc.nih.gov/apps/swarm.html>



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How does Slurm work internally?

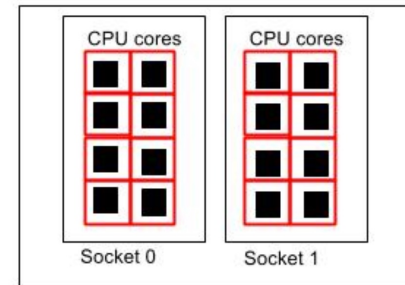
- Bulma request 8 cores from Node 1



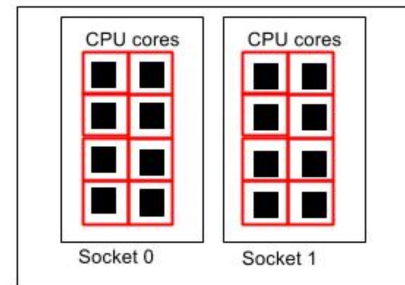
Slurm
scheduler



Node 1

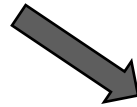


Node 2

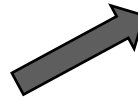


How does Slurm work internally?

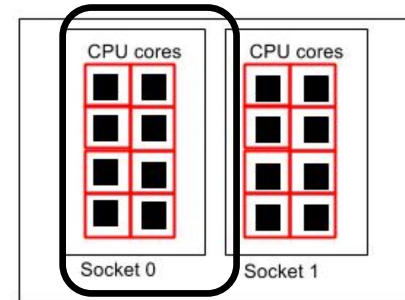
- Bulma request **8 cores** from Node 1



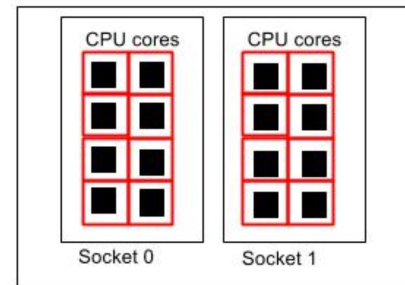
Slurm
scheduler



Node 1

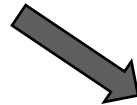


Node 2

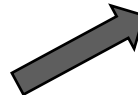


How does Slurm work internally?

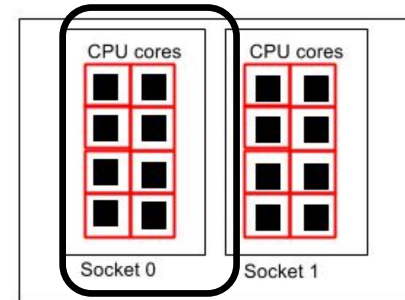
- Bulma request 8 cores from Node 1
- Dr. Brief requests **8 cores** as well from Node 1
- Gohan requests 16 cores from Node 2
- No tragedy of the common!!!



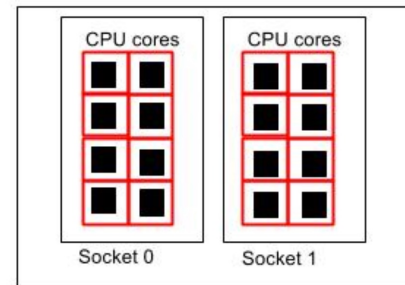
Slurm
scheduler



Node 1

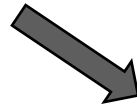


Node 2

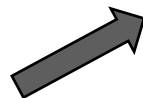


How does Slurm work internally?

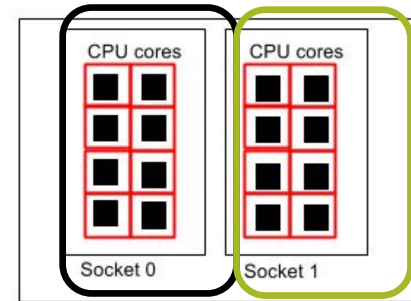
- Bulma request 8 cores from Node 1
- Dr. Brief requests **8 cores** as well from Node 1



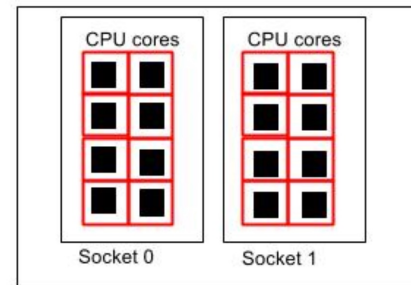
Slurm
scheduler



Node 1

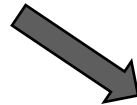


Node 2

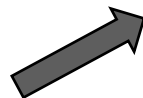


How does Slurm work internally?

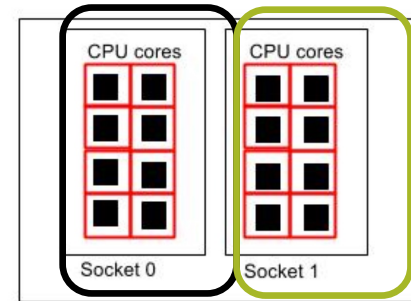
- Bulma request 8 cores from Node 1
- Dr. Brief requests 8 cores as well from Node 1
- Gohan requests **16 cores** from Node 2



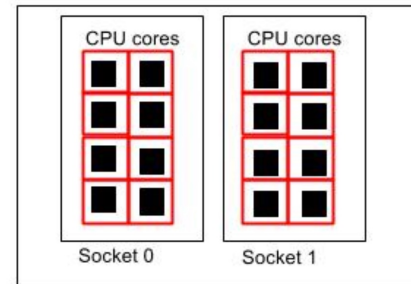
Slurm
scheduler



Node 1

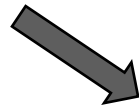


Node 2



How does Slurm work internally?

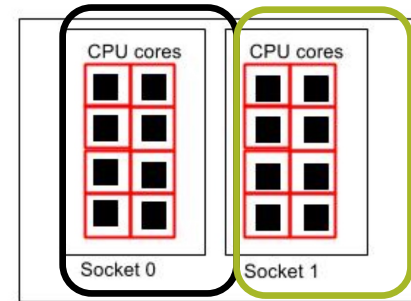
- Bulma request 8 cores from Node 1
- Dr. Brief requests 8 cores as well from Node 1
- Gohan requests **16 cores** from Node 2
- No tragedy of the common!!!



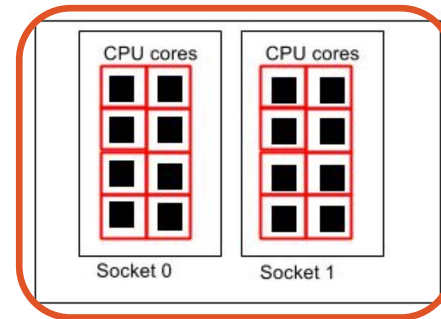
Slurm
scheduler



Node 1



Node 2



Build a Slurm script

- Create the empty slurm script and modify it with your editor of choice (nano, vim, emacs, Ondemand etc ...)



```
[[ @xsede.org@login-ci2 openmpi_hostname]$ touch hello_mpi.sh
[ @xsede.org@login-ci2 openmpi_hostname]$ █
```



Build a Slurm script

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
```



Main CPU
partition name

```
#SBATCH --job-name=first_mpirun
```

```
#SBATCH --output=first_mpirun.%j.out
```

```
#SBATCH --error=first_mpirun.%j.err
```

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@cuanschultz.edu
```

```
#SBATCH --time=00:20:00
```

```
module load gcc
```

```
module load openmpi
```

```
mpirun -n $SLURM_NTASKS hostname
```

Partitions in slurm batch mode

- acpu -> CPU
- amem -> High memory
- aa100 -> NVIDIA GPU partition
- al40 -> NVIDIA GPU partition
- ami100 -> AMD GPU partition



Build a Slurm script

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
```

```
#SBATCH --job-name=first_mpirun
```

 Job name

```
#SBATCH --output=first_mpirun.%j.out
```

```
#SBATCH --error=first_mpirun.%j.err
```

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@cuanschutzh.edu
```

```
#SBATCH --time=00:20:00
```

```
module load gcc
```

```
module load openmpi
```

```
mpirun -n $SLURM_NTASKS hostname
```

Build a Slurm script

```
#!/bin/bash
#SBATCH --partition=acpu
#SBATCH --job-name=first_mpirun
#SBATCH --output=first_mpirun.%j.out
#SBATCH --error=first_mpirun.%j.err
#SBATCH --account=amc-general
#SBATCH --qos=cpu-normal
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --mail-type=ALL
#SBATCH --mail-type=mail-user=user@cuanschultz.edu
#SBATCH --time=00:20:00
module load gcc
module load openmpi

mpirun -n $SLURM_NTASKS hostname
```

Job output file. Similar to stdout in linux.



Build a Slurm script

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
```

```
#SBATCH --job-name=first_mpirun
```

```
#SBATCH --output=first_mpirun.%j.out
```



%j for the jobID number

```
#SBATCH --error=first_mpirun.%j.err
```

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@cuanschultz.edu
```

```
#SBATCH --time=00:20:00
```

```
module load gcc
```

```
module load openmpi
```

```
mpirun -n $SLURM_NTASKS hostname
```

Build a Slurm script

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
```

```
#SBATCH --job-name=first_mpirun
```

```
#SBATCH --output=first_mpirun.%j.out
```

```
#SBATCH --error=first_mpirun.%j.err
```



Job error file.

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@cuanschultz.edu
```

```
#SBATCH --time=00:20:00
```

```
module load gcc
```

```
module load openmpi
```

```
mpirun -n $SLURM_NTASKS hostname
```

Build a Slurm script

```
#!/bin/bash
#SBATCH --partition=acpu
#SBATCH --job-name=first_mpirun
#SBATCH --output=first_mpirun.%j.out
#SBATCH --error=first_mpirun.%j.err
#SBATCH --account=amc-general
#SBATCH --qos=cpu-normal
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --mail-type=ALL
#SBATCH --mail-type=mail-user=user@cuanschutzh.edu
#SBATCH --time=00:20:00
module load gcc
module load openmpi

mpirun -n $SLURM_NTASKS hostname
```



Account name. All AMC users should use the **amc-general** account!

Build a Slurm script

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
```

```
#SBATCH --job-name=first_mpirun
```

```
#SBATCH --output=first_mpirun.%j.out
```

```
#SBATCH --error=first_mpirun.%j.err
```

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
```



Quality of service

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@cuanschutzu.edu
```

```
#SBATCH --time=00:20:00
```

```
module load gcc
```

```
module load openmpi
```

```
mpirun -n $SLURM_NTASKS hostname
```

Slurm Quality of service (qos)

- Used to modify or constrain characteristics that a job can have.
- It is now mandatory to use **#SBATCH --qos** when interacting with Slurm!



QOS changes on Alpine

- To improve resource availability across Alpine partitions and simplify job submissions, several QOS changes have been made.



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QOS changes on Alpine

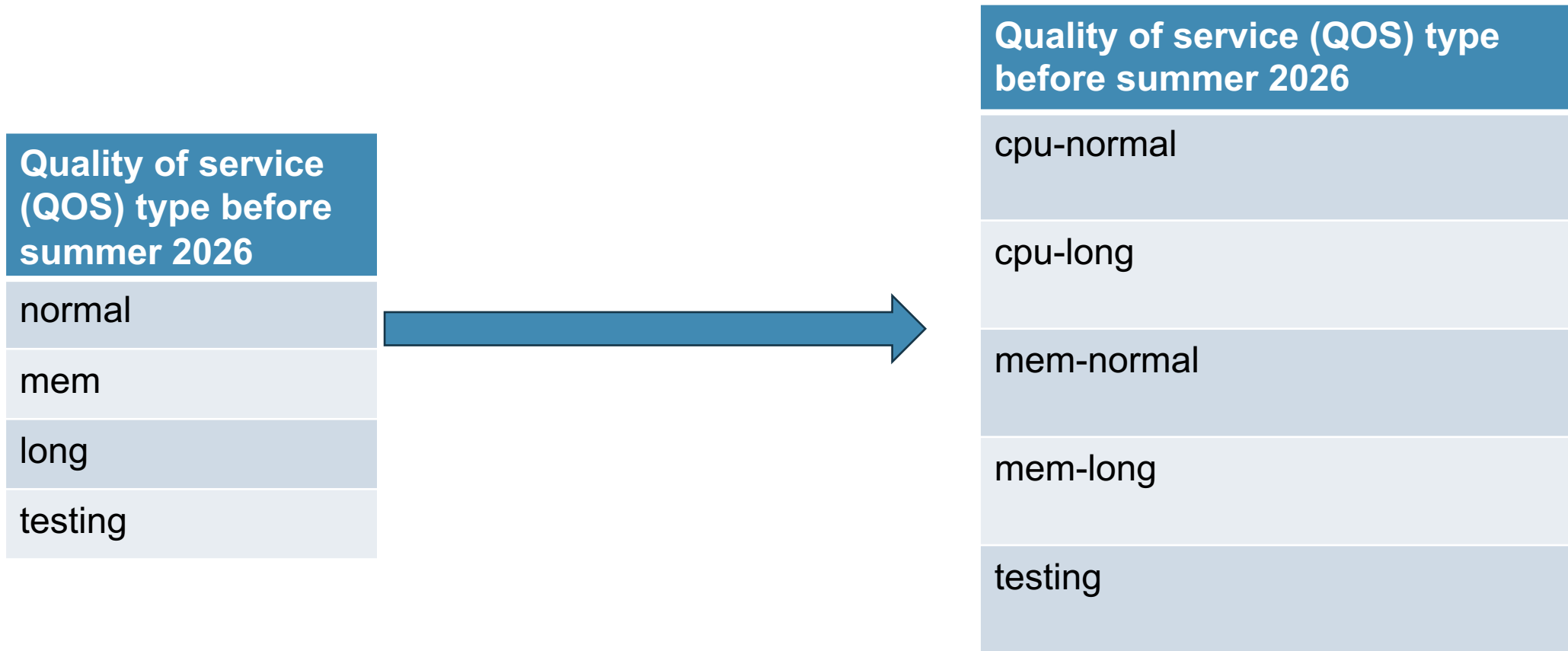
- To improve resource availability across Alpine partitions and simplify job submissions, several QOS changes have been made.
- The new QOS combines the old qos definition with the generic resource (GRES) type in order to reduce very long wait times.



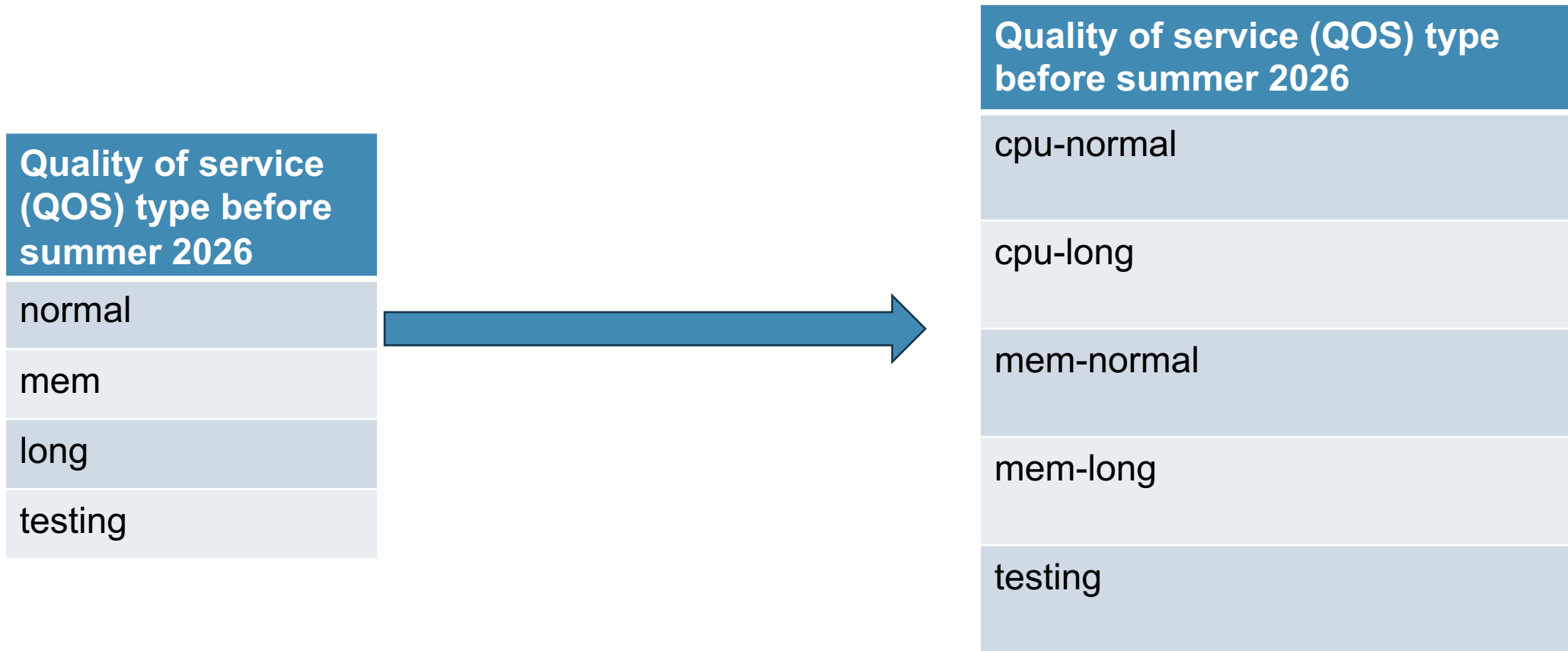
Slurm Quality of service (qos)

Quality of service (QOS) type	Quality of service (QOS) type
cpu-normal	Corresponds to jobs submitted to the acpu partition under 24 hours.
cpu-long	Corresponds to jobs submitted to the acpu partition over 24 hours.
mem-normal	Corresponds to jobs submitted to the amem partition under 24 hours.
mem-long	Corresponds to jobs submitted to the amem partition over 24 hours.
testing	Corresponds to jobs submitted to the atesting partition, typically in as interactive. The walltime cannot exceed 1 hour

Summary of the changes



Summary of the changes



Build a Slurm script

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
```

```
#SBATCH --job-name=first_mpirun
```

```
#SBATCH --output=first_mpirun.%j.out
```

```
#SBATCH --error=first_mpirun.%j.err
```

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
```

```
#SBATCH --nodes=1
```



Number of nodes. **Always use 1** unless you are using MPI, gnu parallel or sparkcluster!

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@cuanschutzh.edu
```

```
#SBATCH --time=00:20:00
```

```
module load gcc
```

```
module load openmpi
```

```
mpirun -n $SLURM_NTASKS hostname
```

Build a Slurm script

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
```

```
#SBATCH --job-name=first_mpirun
```

```
#SBATCH --output=first_mpirun.%j.out
```

```
#SBATCH --error=first_mpirun.%j.err
```

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=4
```



Number of CPU cores requested.

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@cuanschultz.edu
```

```
#SBATCH --time=00:20:00
```

```
module load gcc
```

```
module load openmpi
```

```
mpirun -n $SLURM_NTASKS hostname
```

Build a Slurm script

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
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#SBATCH --job-name=first_mpirun
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#SBATCH --output=first_mpirun.%j.out
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#SBATCH --error=first_mpirun.%j.err
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#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
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```
#SBATCH --nodes=1
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```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@cuanschutzh.edu
```

```
#SBATCH --time=00:20:00
```

```
module load gcc
```

```
module load openmpi
```

```
mpirun -n $SLURM_NTASKS hostname
```

To receive updates about the slurm jobs, including on CPU and memory efficiency.



Email Update from the slurm job

alpine **Slurm** Job_id=19861317 Name=build

Ended, Run time 00:33:51, COMPLETED, ExitCode 0



slurm@colorado.edu

To:

Job ID: 19861317

Cluster: alpine

User/Group:

State: COMPLETED (exit code 0)

Nodes: 1

Cores per node: 8

CPU Utilized: 03:42:24

CPU Efficiency: 82.13% of 04:30:48 core-walltime

Job Wall-clock time: 00:33:51

Memory Utilized: 5.30 GB

Memory Efficiency: 17.67% of 30.00 GB (3.75 GB/core)



>80% of job efficiency is acceptable but ideally we would like either a CPU or memory efficiency to be above 90%.



Slurm example on number of cores

- hostname is just a program that prints the name of the node I am working with



```
[[@xsede.org@c3cpu-a5-u32-1 ~]$ mpirun -n 4 hostname  
c3cpu-a5-u32-1.rc.int.colorado.edu  
c3cpu-a5-u32-1.rc.int.colorado.edu  
c3cpu-a5-u32-1.rc.int.colorado.edu  
c3cpu-a5-u32-1.rc.int.colorado.edu
```



- With mpirun, we schedule 4 tasks on the 4 cores,
thus the name of the node gets printed 4 times !!!

Build a Slurm script

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
```

```
#SBATCH --job-name=first_mpirun
```

```
#SBATCH --output=first_mpirun.%j.out
```

```
#SBATCH --error=first_mpirun.%j.err
```

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@cuanschultz.edu
```

```
#SBATCH --time=00:20:00
```



Duration of the slurm job.

```
module load gcc
```

```
module load openmpi
```

```
mpirun -n $SLURM_NTASKS hostname
```

Build a Slurm script

```
#!/bin/bash
#SBATCH --partition=acpu
#SBATCH --job-name=first_mpirun
#SBATCH --output=first_mpirun.%j.out
#SBATCH --error=first_mpirun.%j.err
#SBATCH --account=amc-general
#SBATCH --qos=cpu-normal
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --mail-type=ALL
#SBATCH --mail-type=mail-user=user@cuanschutz.edu
#SBATCH --time=00:20:00
module load gcc
module load openmpi
mpirun -n $SLURM_NTASKS hostname
```



Packages are not loaded automatically! You need to always load them.

Build a Slurm script

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
```

```
#SBATCH --job-name=first_mpirun
```

```
#SBATCH --output=first_mpirun.%j.out
```

```
#SBATCH --error=first_mpirun.%j.err
```

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=
```

```
#SBATCH --mail-type=mail-user=...@cuanschutz.edu
```

```
#SBATCH --time=00:20:00
```

```
module load gcc
```

```
module load openmpi
```

```
mpirun -n $SLURM_NTASKS hostname
```

\$SLURM_NTASKS is the slurm env variable
for the 4 cores

Recap

```
#!/bin/bash
```

```
#SBATCH --partition=amilan
```

```
#SBATCH --job-name=first_mpirun
```

```
#SBATCH --output=first_mpirun.%j.out
```

```
#SBATCH --error=first_mpirun.%j.err
```

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=normal
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@
```

```
cuanschultz.edu
```

```
#SBATCH --time=00:20:00
```



```
#!/bin/bash
```

```
#SBATCH --partition=acpu
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```

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@
```

```
cuanschultz.edu
```

```
#SBATCH --time=00:20:00
```

Build a Slurm script

We use sbatch to submit the script



```
[[      @xsede.org@login-ci2 openmpi_hostname]$ sbatch hello_mpi.sh  
Submitted batch job 2998325
```

Build a Slurm script

We get output and error files, ending with %j.out and %j.err



```
(sede.org@login-ci2 openmpi_hostname]$ ls -latr
total 184
drwxrws---. 44          1672 Sep 15 01:45 ..
-rw-r--r--.  1          400 Sep 15 01:52 hello_mpi.sh
-rw-r--r--.  1           0 Sep 15 01:52 first_mpirun-job.2998325.err
drwxr-sr-x.  2          122 Sep 15 01:52 .
-rw-r--r--.  1          140 Sep 15 01:52 first_mpirun-job.2998325.out
(sede.org@login-ci2 openmpi_hostname]$ cat first_mpirun-job.2998325.out
c3cpu-c15-u1-2.rc.int.colorado.edu
c3cpu-c15-u1-2.rc.int.colorado.edu
c3cpu-c15-u1-2.rc.int.colorado.edu
c3cpu-c15-u1-2.rc.int.colorado.edu
(sede.org@login-ci2 openmpi_hostname]$
```





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Best practices

Alpine status update

- Please refer or sign up to the following website to get updates regarding Alpine status: <https://curc.statuspage.io/>



SUBSCRIBE TO UPDATES

All Systems Operational

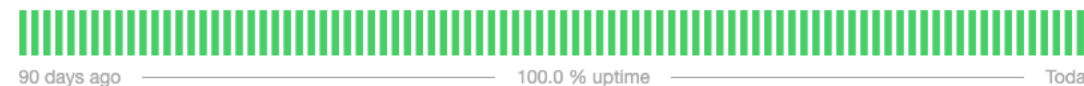
Uptime over the past 90 days. [View historical uptime.](#)

Research Computing Core ?

Operational

Alpine ?

Operational



Blanca ?

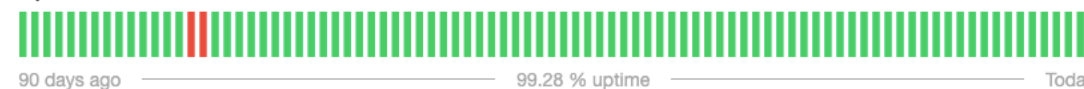
Operational

PetaLibrary

Operational

Open OnDemand ?

Operational



Alpine job arrays limit

- In order to ensure accessibility and fairness, job arrays are limited to a maximum of 200 concurrent jobs that can be submitted at the same time:
<https://curc.readthedocs.io/en/latest/clusters/alpine/alpine-hardware.html#cu-anschutz-medical-campus-contribution>
- Please consider using alternatives if needed such as gnu parallel or emailing rc-help@colorado.edu.

Persistence1

- If you are using nextflow or snakemake and need to run your jobs for a long time (>24 hours), consider emailing rc-help@colorado.edu to get access to Persistence1.
- For more information regarding Persistence1 please refer to this page: <https://curc.readthedocs.io/en/latest/running-jobs/persistence1.html#running-workflow-managers-on-persistence1> .



Alpine data transfer

- If you need access to ssh so that you can use scp, sftp or rsync, Open a ticket and fill out this form:
https://ucdenverdata.formstack.com/forms/alpine_ssh_request_form
- We do one onboarding session per month.



How to estimate my GPU core hours?

- We need to be mindful of the cpu resources that we request on Alpine.
- Our campus has a total of 6,459 total shares which means that we can spend about 6,459 cores per hour in total before our priority starts to get affected.
- Let's see a scenario that highlights the cpu core hours calculation.

Scenario

- User A requests 8 cpu ntasks for 1 hour
- The formula goes as follows:

Total SU = [(total_cores * TRESBillingWeightsCPU) + (total_cpu_mem * TRESBillingWeightsMem)] * total_hours

Scenario

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- The formula goes as follows:

$$\text{Total SU} = [(\text{total_cores} * \text{TRESBillingWeightsCPU}) + (\text{total_cpu_mem} * \text{TRESBillingWeightsMem})] * \text{total_hours}$$

$$\text{Total SU} = [(8 * 0.5) + (28.31 \text{ GiB} * 0.141)] * 1 \text{ hour}$$

Scenario

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- The formula goes as follows:

$$\text{Total SU} = [(\text{total_cores} * \text{TRESBillingWeightsCPU}) + (\text{total_cpu_mem} * \text{TRESBillingWeightsMem})] * \text{total_hours}$$

$$\begin{aligned}\text{Total SU} &= [(8 * 0.5) + (28.31 \text{ GiB} * 0.141)] * 1 \text{ hour} \\ &= [4 + 3.99171] * 1 \text{ hour} \\ &= 7.99171 \text{ SUs or core hours}\end{aligned}$$

Scenario

- You can find all cpu Billingweights here:

<https://curc.readthedocs.io/en/latest/clusters/alpine/alpine-hardware.html#partitions-available-on-alpine>

- The calculation formula comes from this page:
<https://slurm.schedmd.com/tres.html>



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THANK YOU